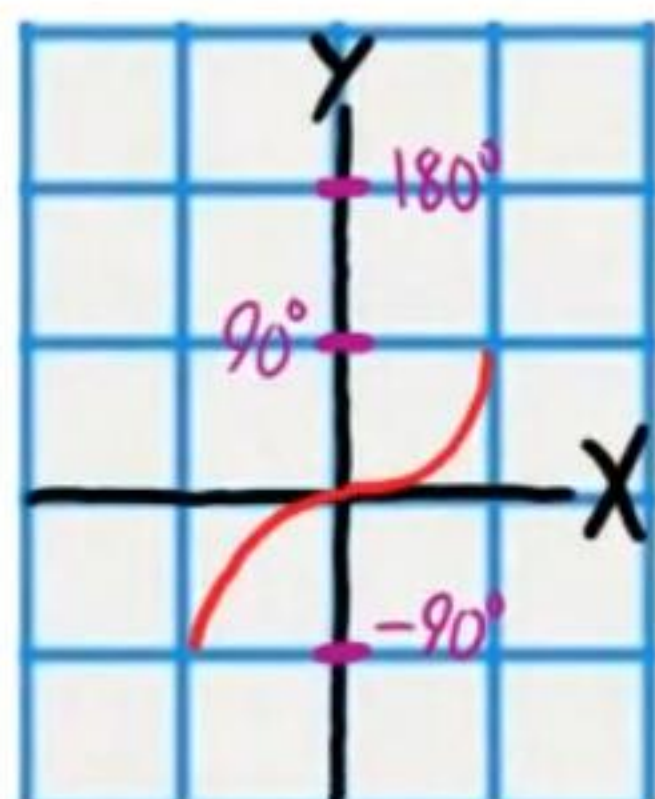


Evaluating Composite Trigonometric Functions

Inverse Trigonometric Functions and Their Graphs

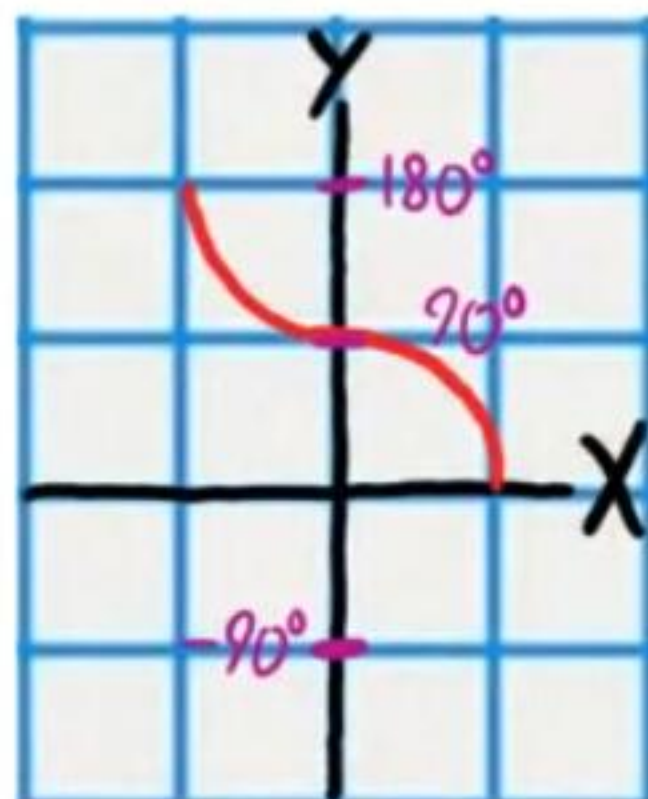
$$y = \sin^{-1}x$$

Range: $-90^\circ \leq y \leq 90^\circ$



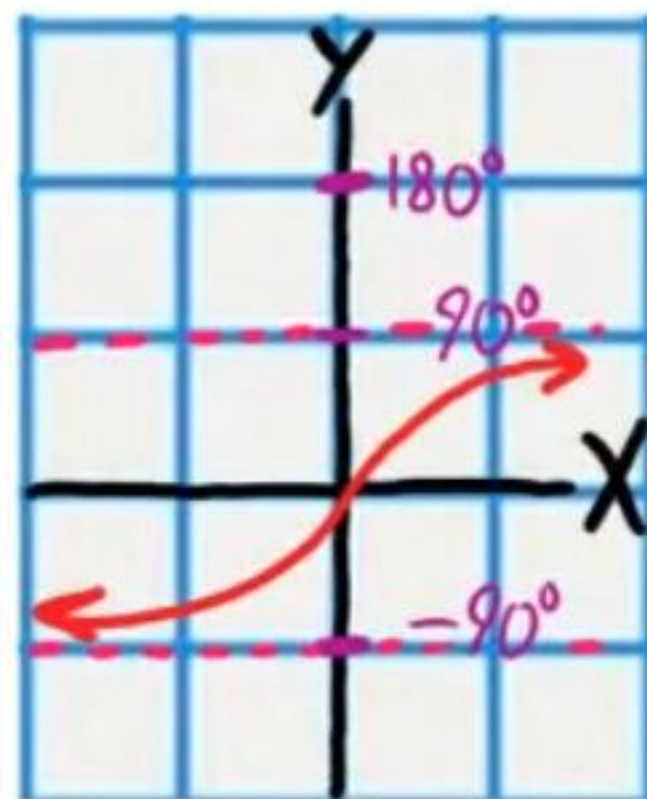
$$y = \cos^{-1}x$$

$0^\circ \leq y \leq 180^\circ$



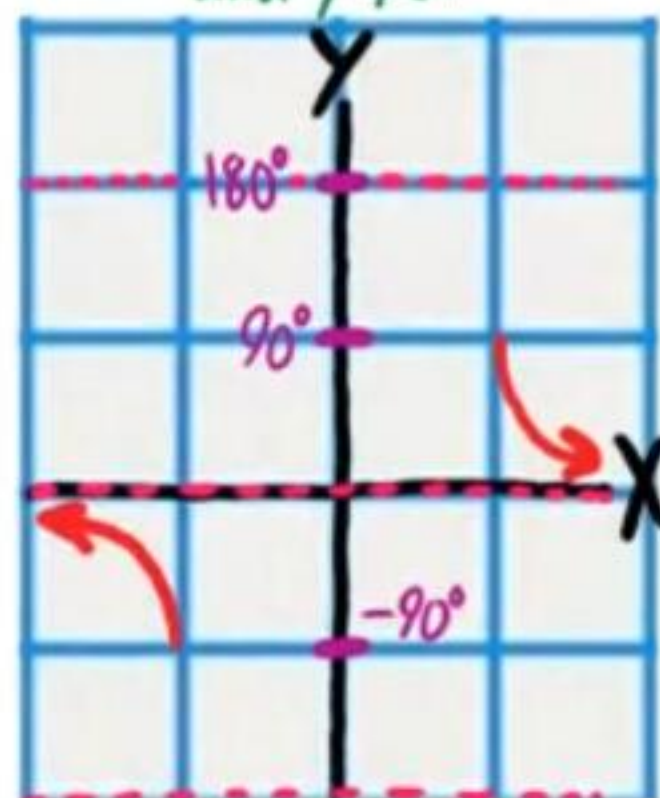
$$y = \tan^{-1}x$$

$-90^\circ < y < 90^\circ$



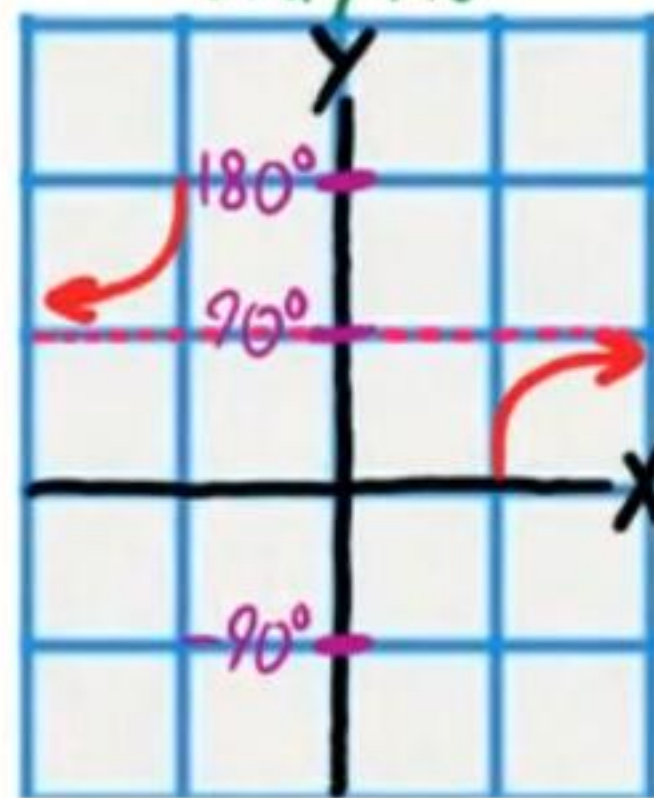
$$y = \csc^{-1}x$$

Range: $-90^\circ \leq y \leq 90^\circ$
and $y \neq 0$.



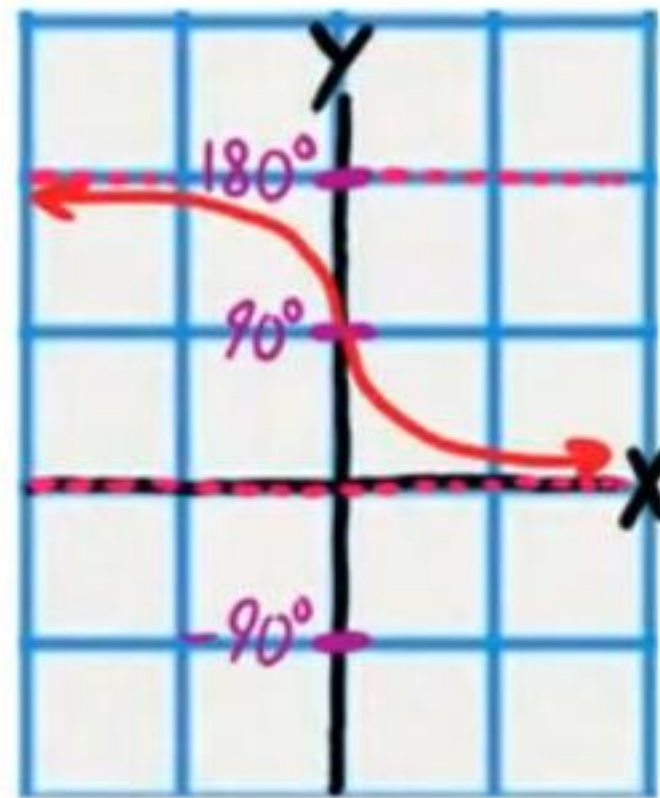
$$y = \sec^{-1}x$$

$0^\circ \leq y \leq 180^\circ$
and $y \neq 90^\circ$



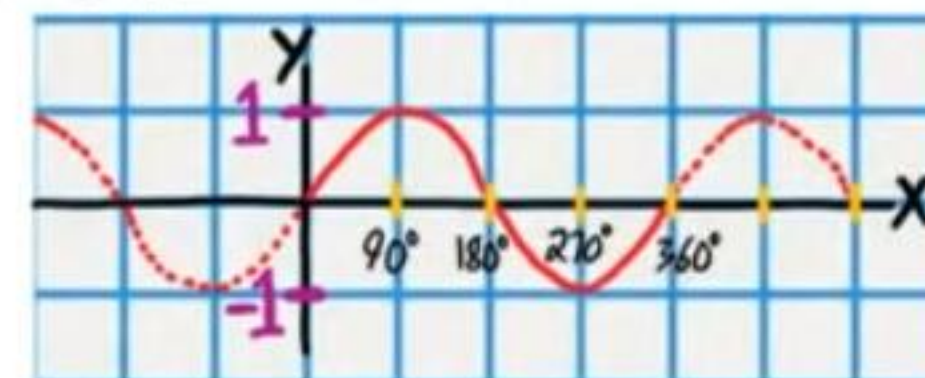
$$y = \cot^{-1}x$$

$0^\circ < y < 180^\circ$

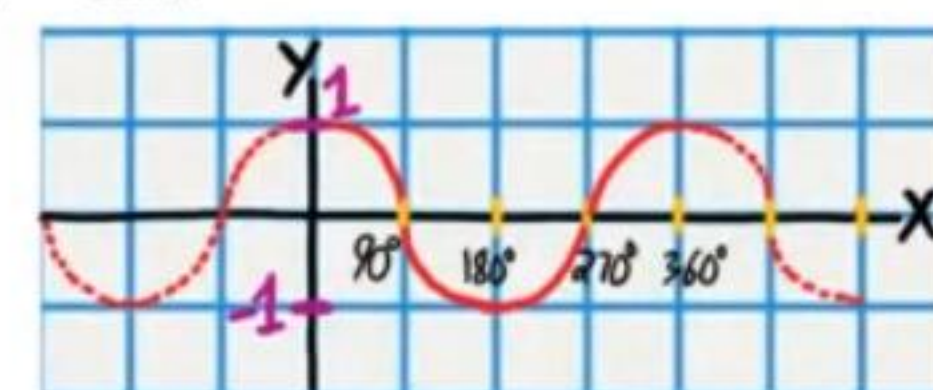


The Graphs of Basic Trigonometric Functions

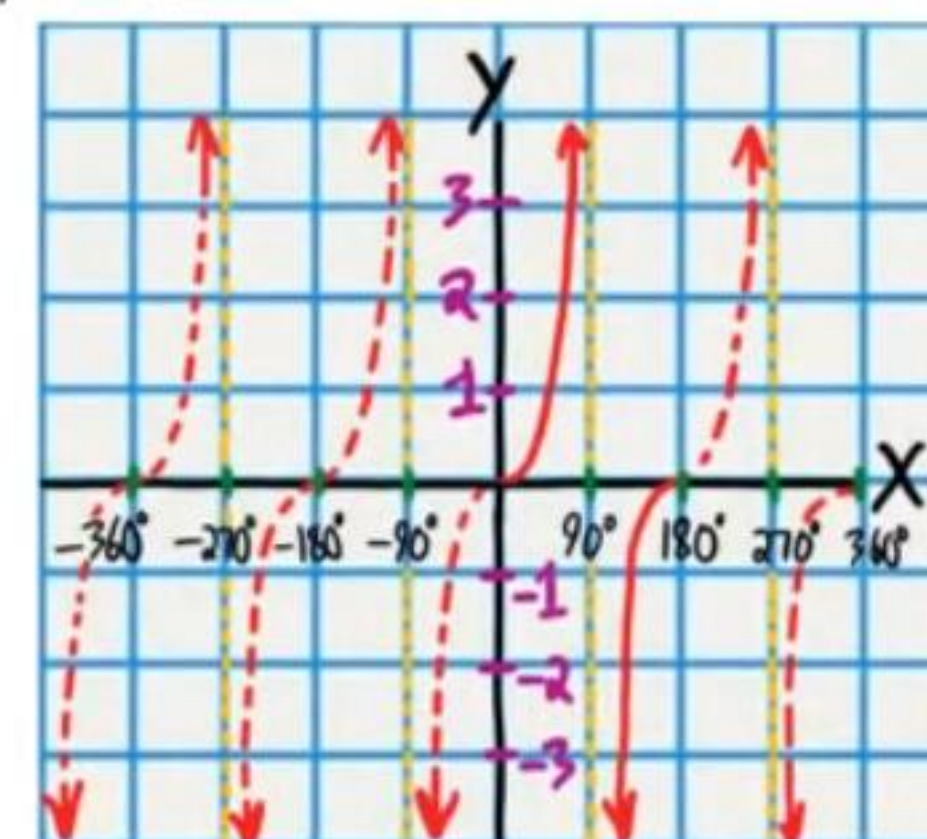
$$y = \sin x$$



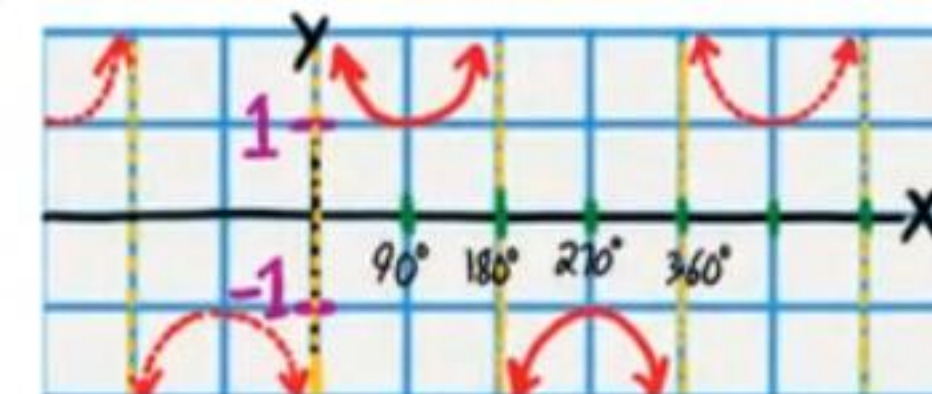
$$y = \cos x$$



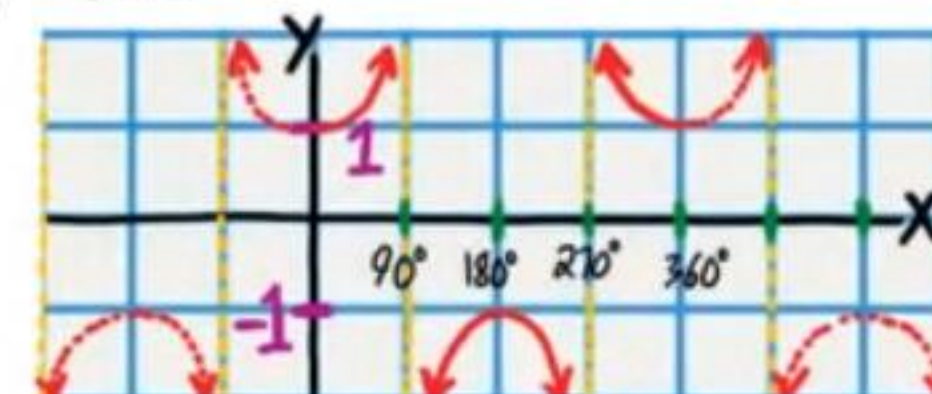
$$y = \tan x$$



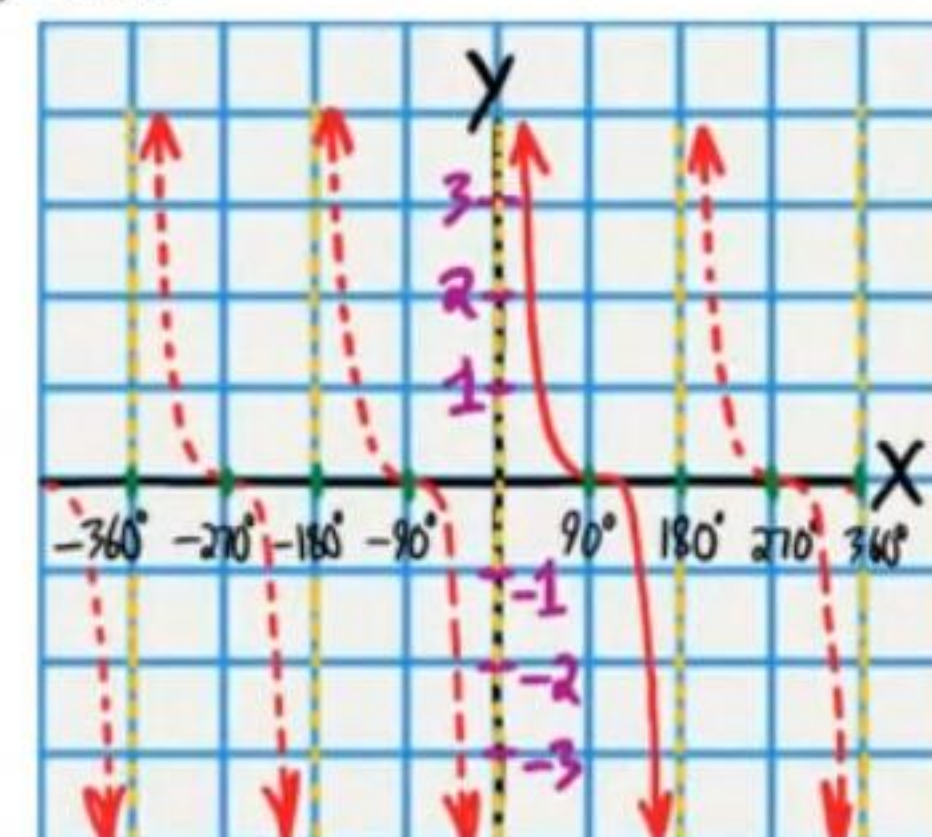
$$y = \csc x$$



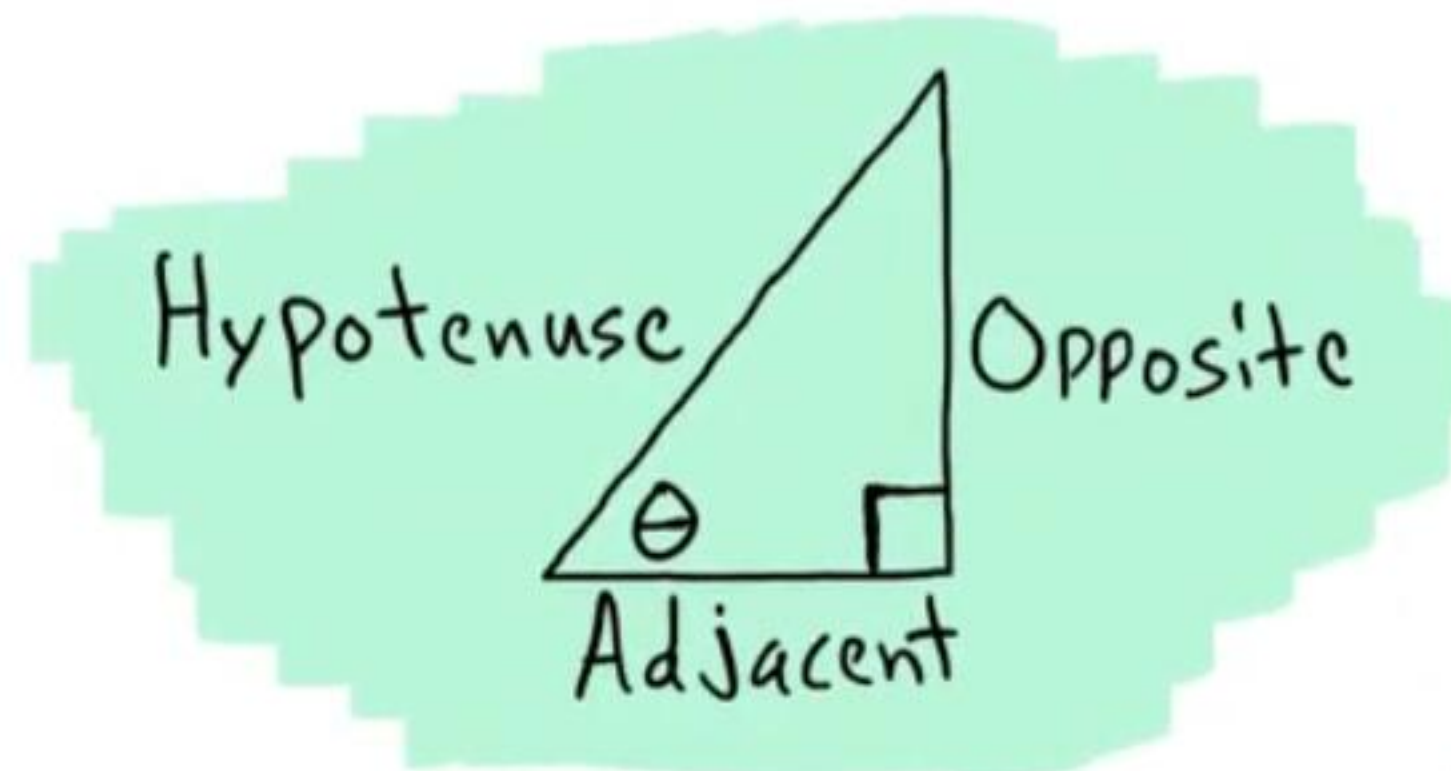
$$y = \sec x$$



$$y = \cot x$$



The Six Basic Trigonometric Functions



$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

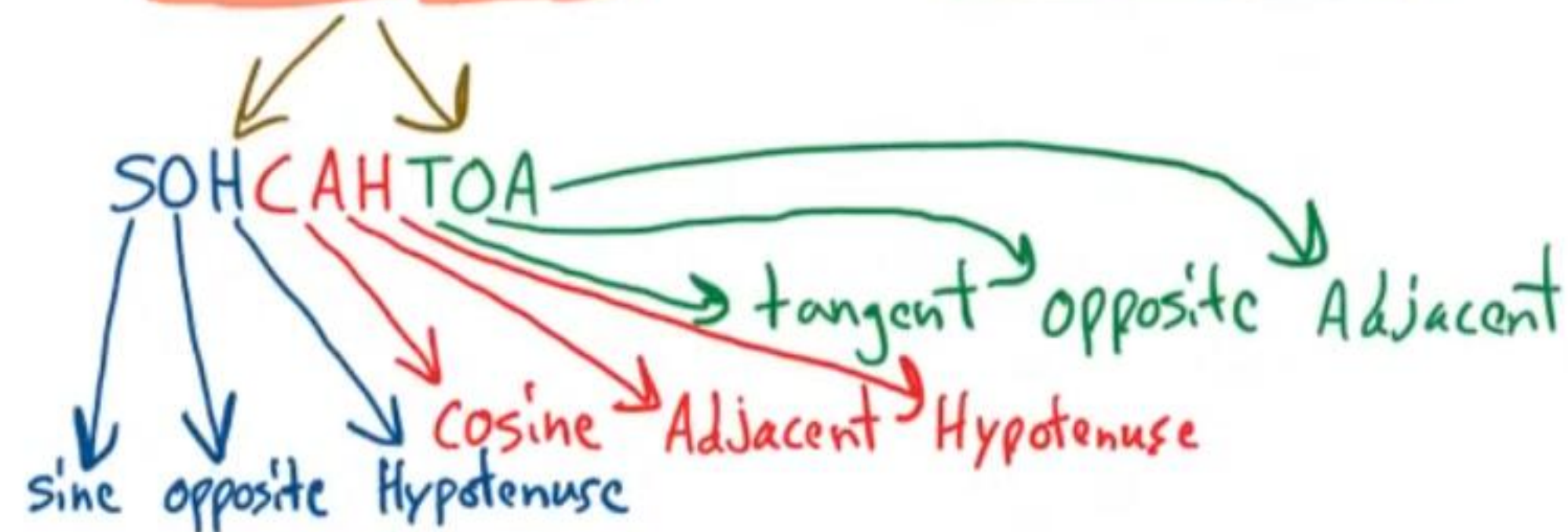
$$\csc \theta = \frac{\text{HYP}}{\text{OPP}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

$$\sec \theta = \frac{\text{HYP}}{\text{Adj}}$$

$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

$$\cot \theta = \frac{\text{Adj}}{\text{OPP}}$$

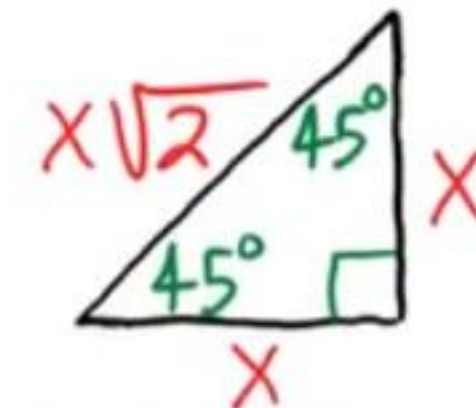


Special Right Triangles

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is X , then the length of the hypotenuse is $2X$ and the length of the side opposite the 60° angle is $X\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are X , then the length of the hypotenuse is $X\sqrt{2}$.



Find the value of each expression *without a calculator*.

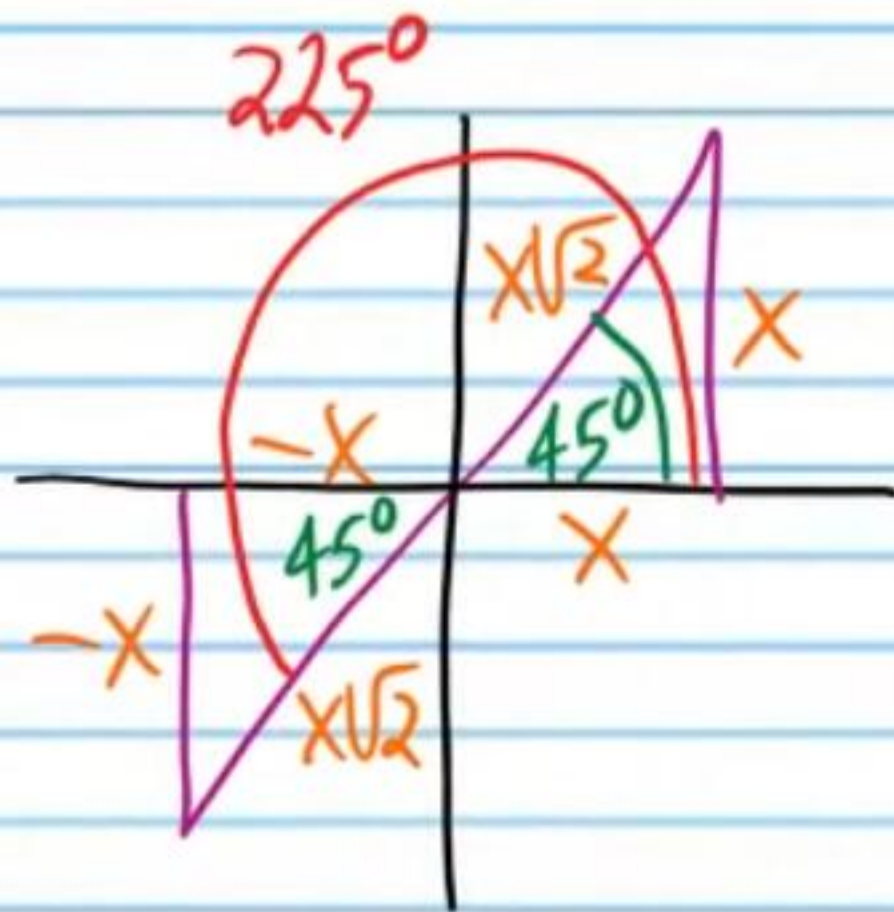
• ① $\sin(\tan^{-1} 1)$

I) $\theta = \tan^{-1} 1$

$\tan \theta = 1$

$\theta = 45^\circ, 225^\circ$

$\tan^{-1} 1 = 45^\circ$



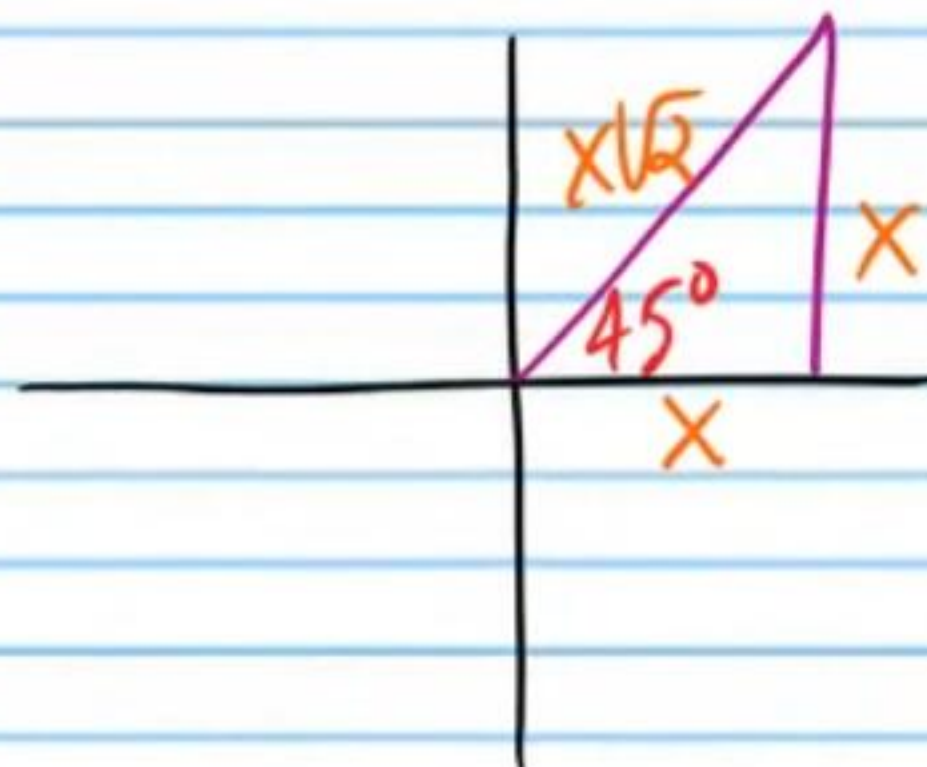
II) $\sin(\tan^{-1} 1)$

$\sin 45^\circ$

$\frac{1}{\sqrt{2}}$

$\frac{1}{\sqrt{2}}$

$\boxed{\frac{\sqrt{2}}{2}}$



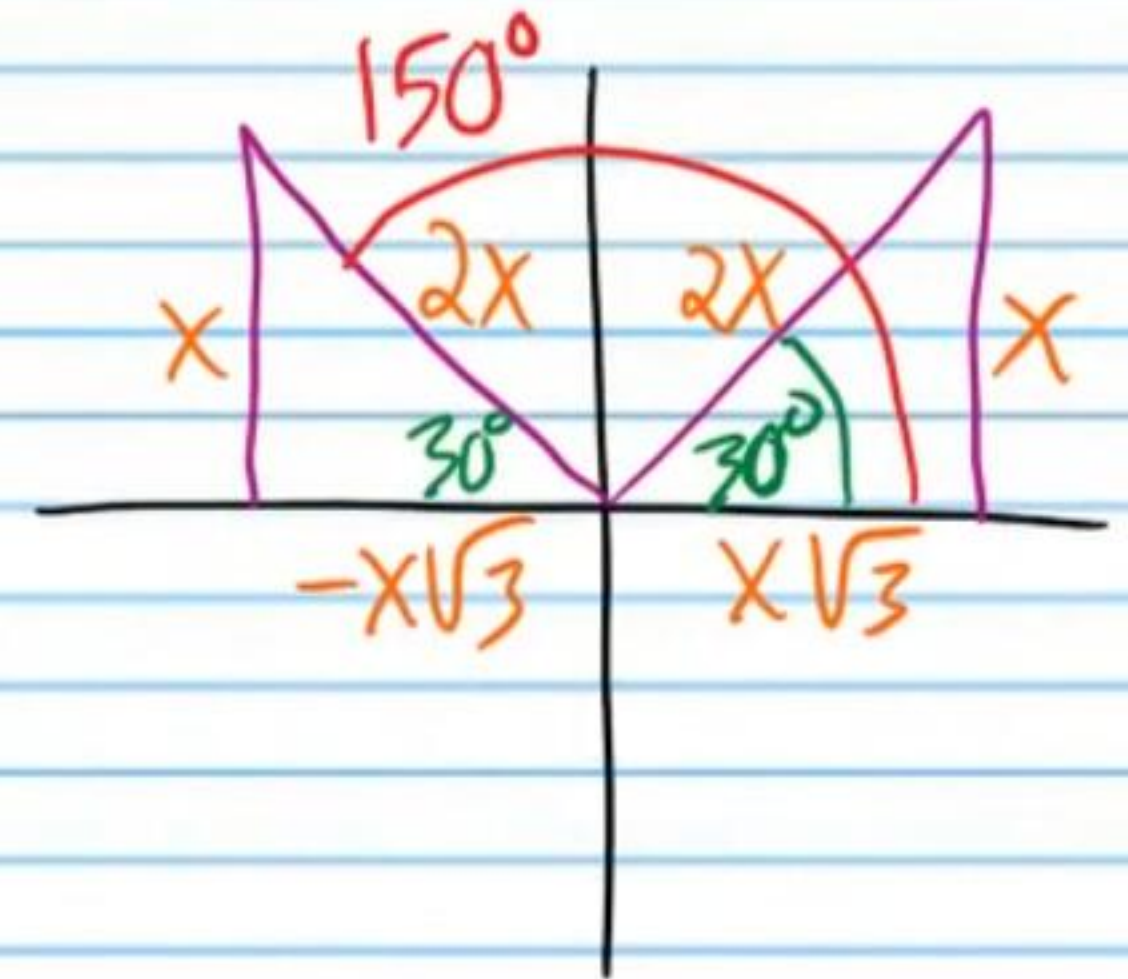
② $\sec(\sin^{-1} \frac{1}{2})$

I) $\theta = \sin^{-1} \frac{1}{2}$

$\sin \theta = \frac{1}{2}$

$\theta = 30^\circ, 150^\circ$

$\sin^{-1} \frac{1}{2} = 30^\circ$



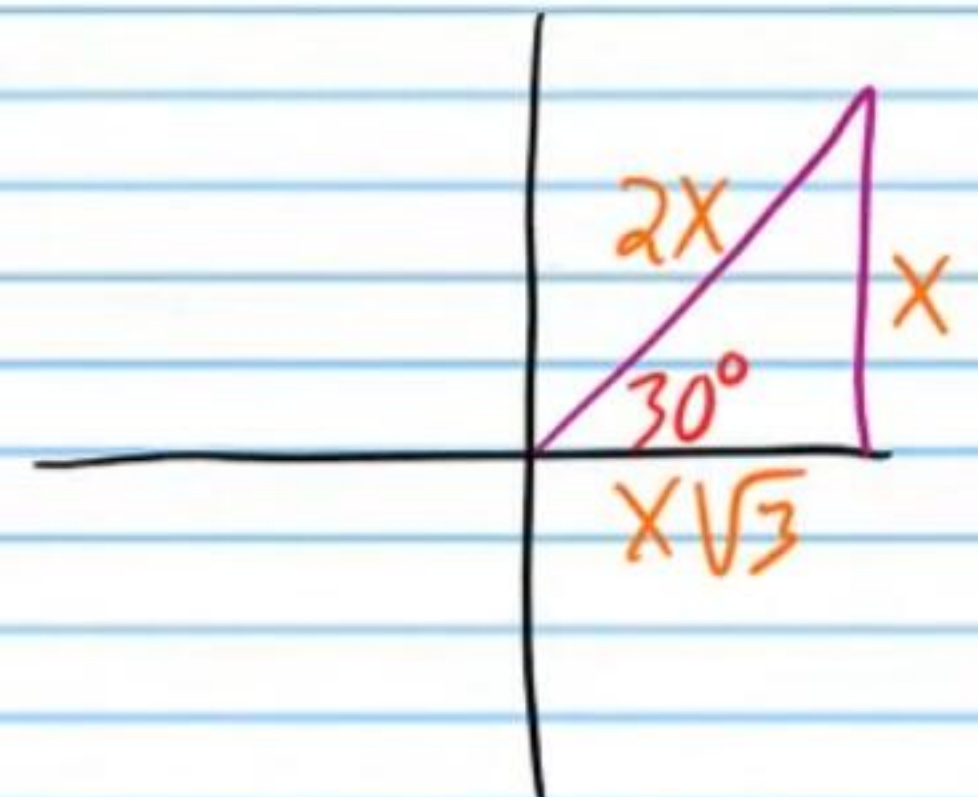
II) $\sec(\sin^{-1} \frac{1}{2})$

$\sec 30^\circ$

$\frac{2}{\sqrt{3}}$

$\frac{2}{\sqrt{3}}$

$\boxed{\frac{2\sqrt{3}}{3}}$



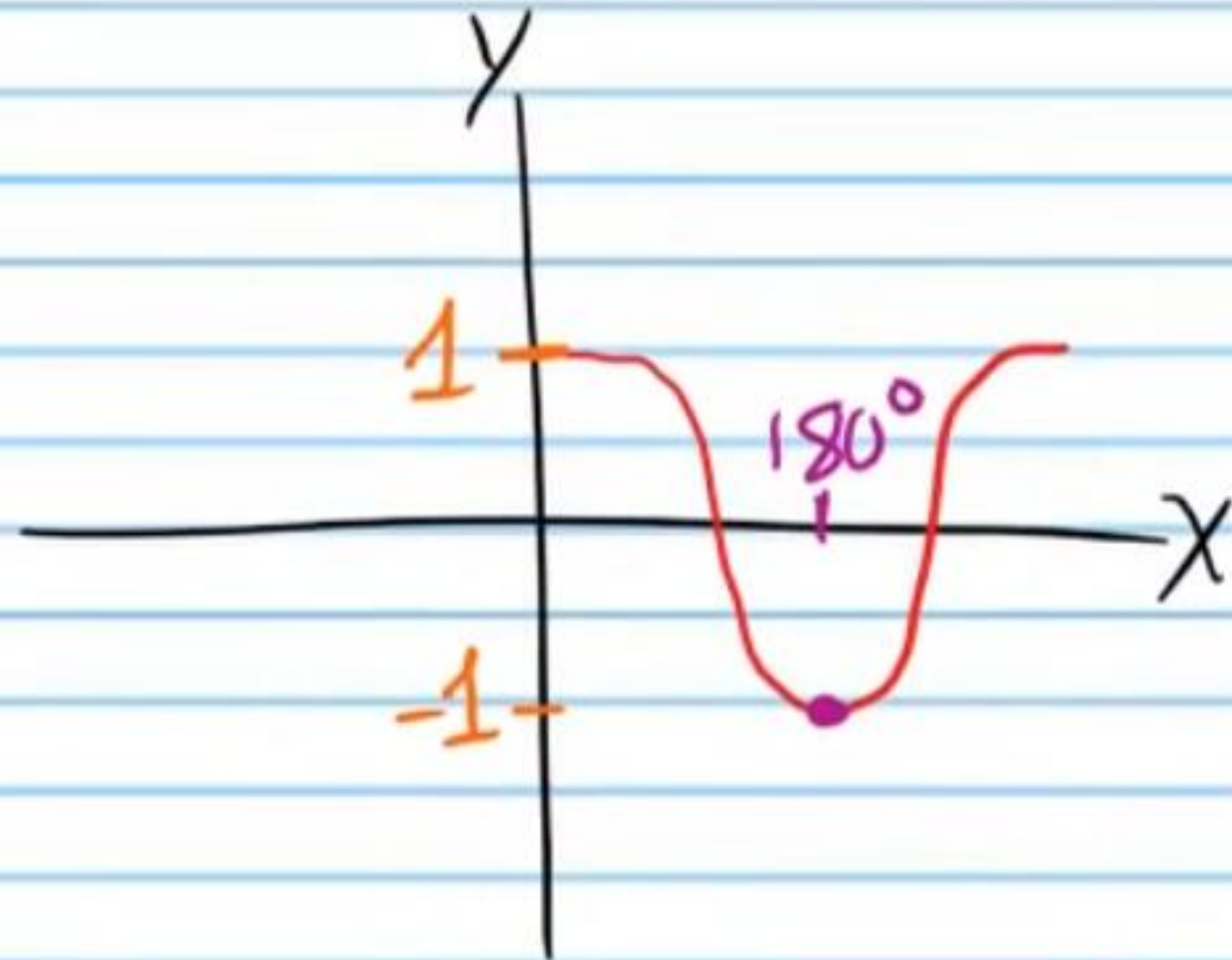
③ $\tan(\cos^{-1}(-1))$

I) $\theta = \cos^{-1}(-1)$

$\cos \theta = -1$

$\theta = 180^\circ$

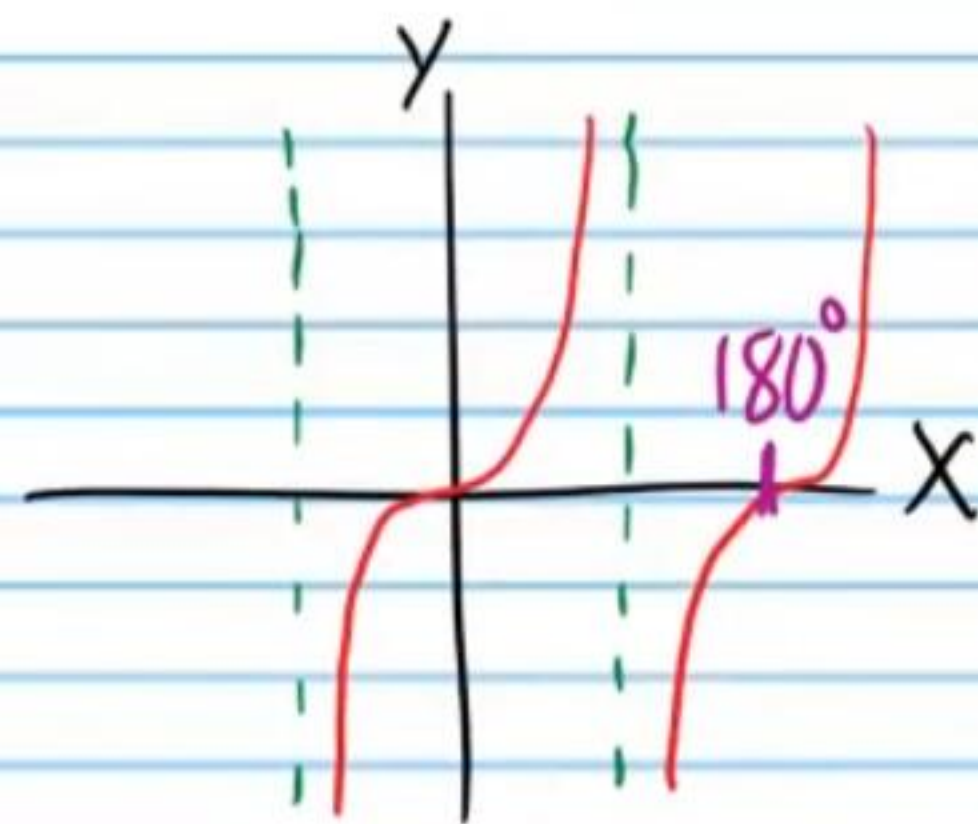
$\cos^{-1}(-1) = 180^\circ$



II) $\tan(\cos^{-1}(-1))$

$\tan 180^\circ$

$\boxed{0}$



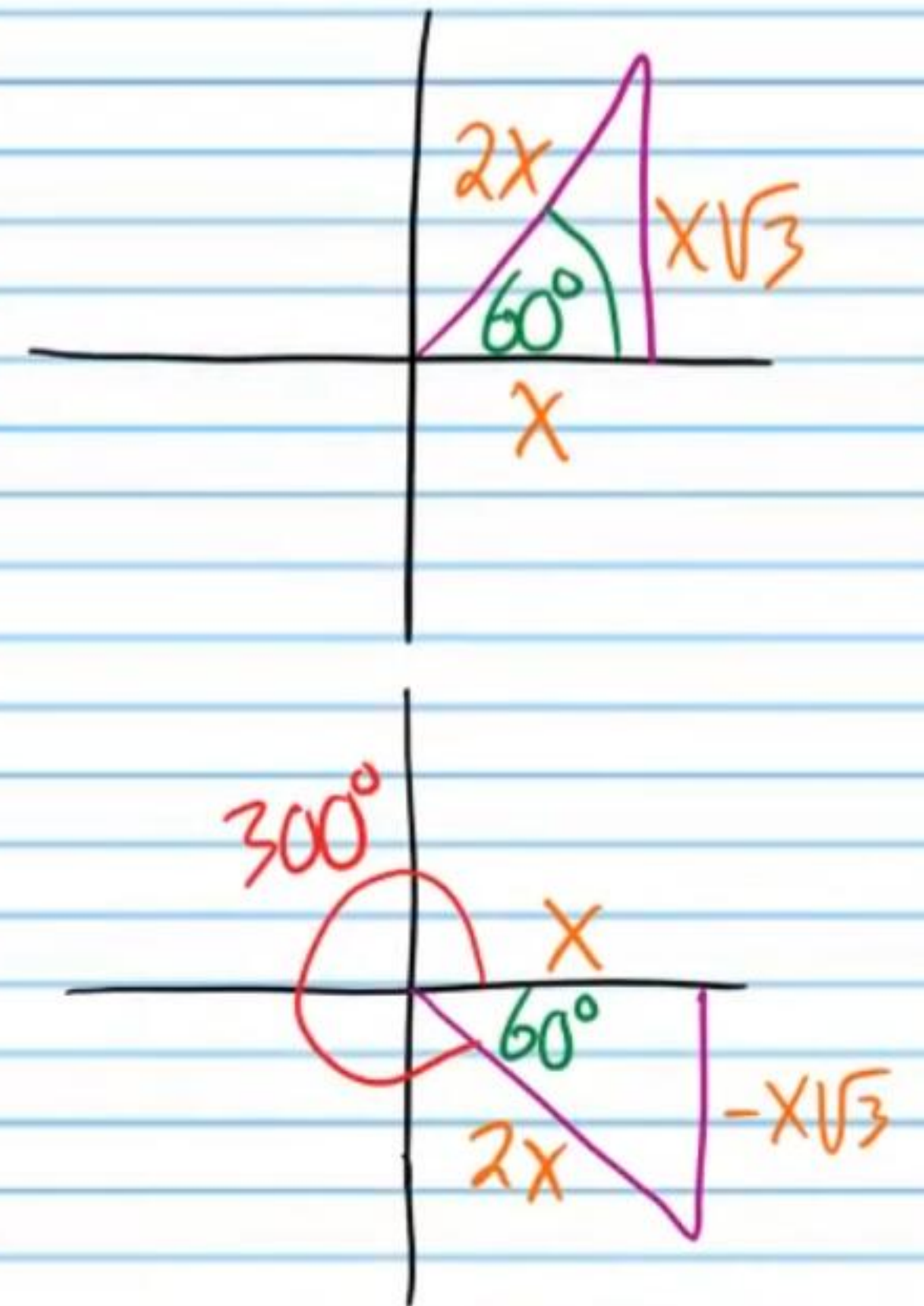
④ $\csc(\sec^{-1}2)$

I) $\theta = \sec^{-1}2$

$\sec \theta = 2$

$\theta = 60^\circ, 300^\circ$

$\sec^{-1}2 = 60^\circ$



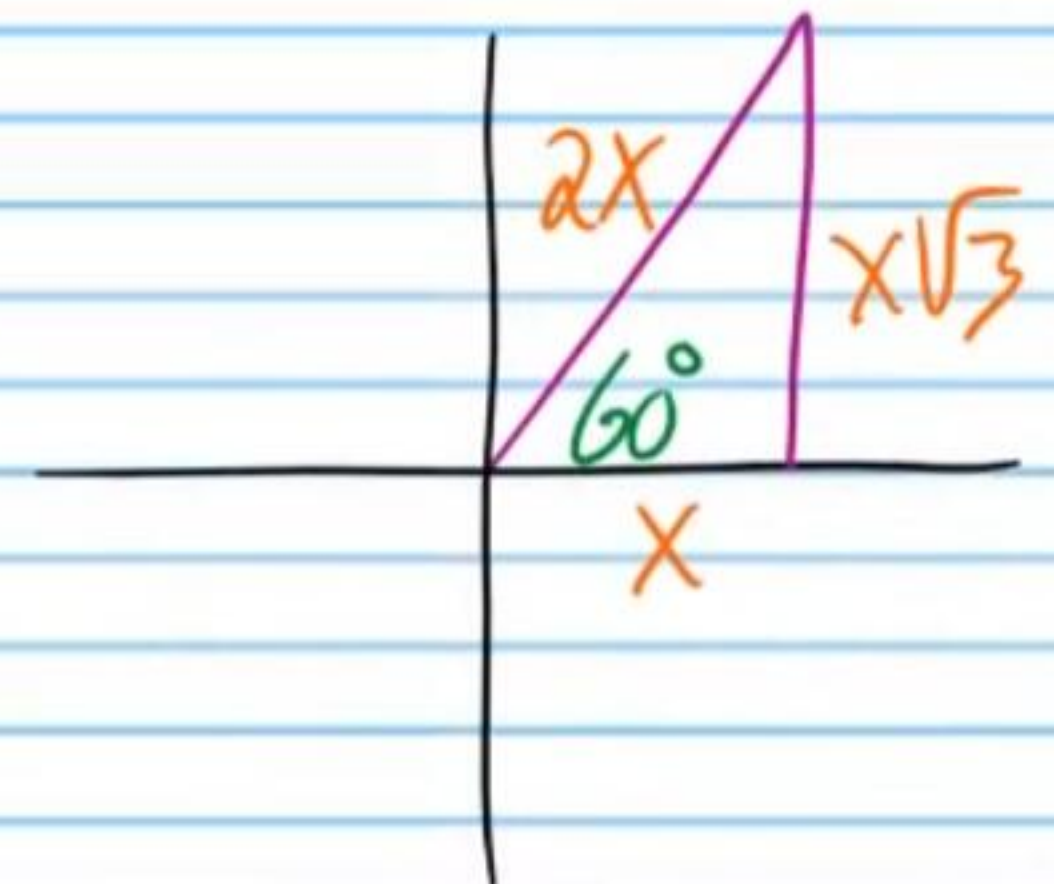
II) $\csc(\sec^{-1}2)$

$\csc 60^\circ$

$\frac{2x}{x\sqrt{3}}$

$\frac{2}{\sqrt{3}}$

$\boxed{\frac{2\sqrt{3}}{3}}$



⑤ $\csc(\sin^{-1} \frac{1}{\sqrt{2}})$

I) $\theta = \sin^{-1} \frac{1}{\sqrt{2}}$

$$\sin \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ, 135^\circ$$

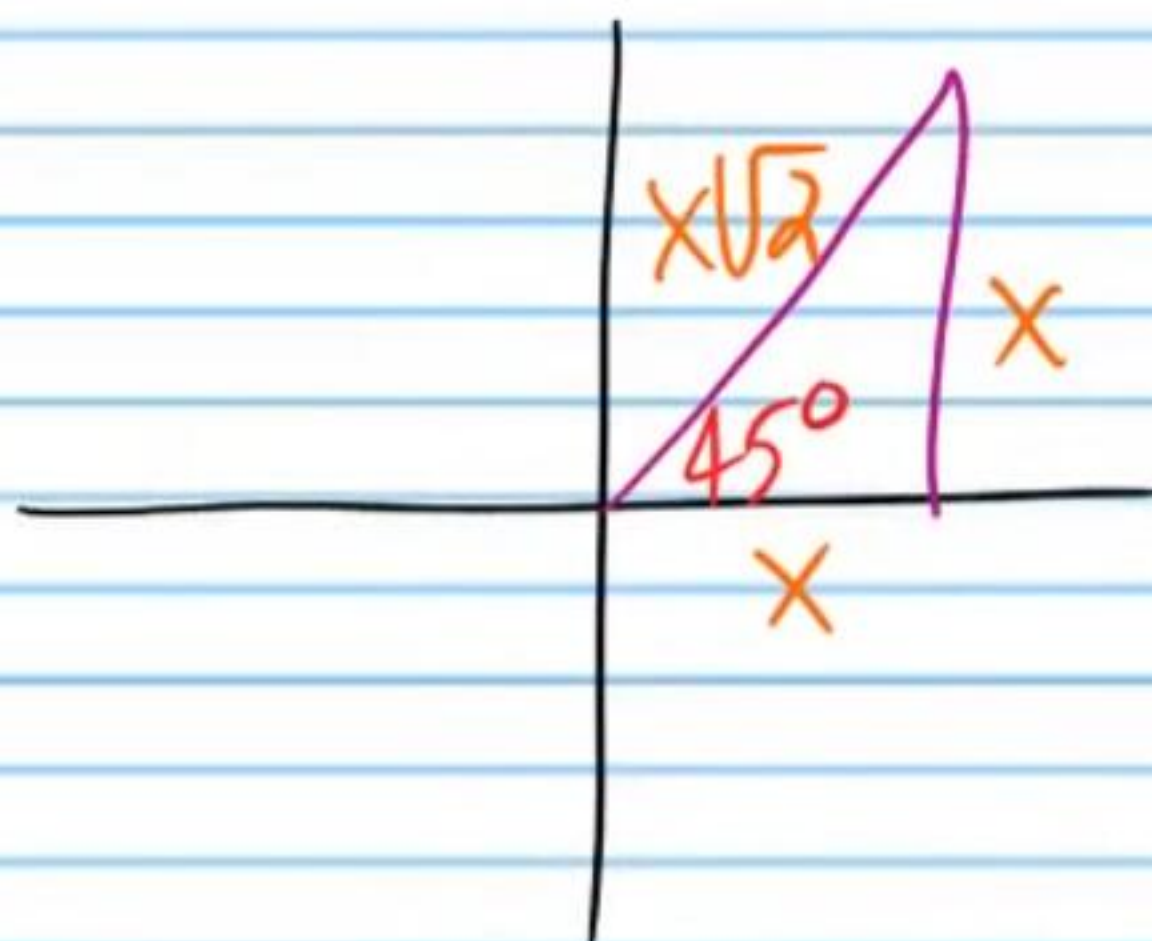
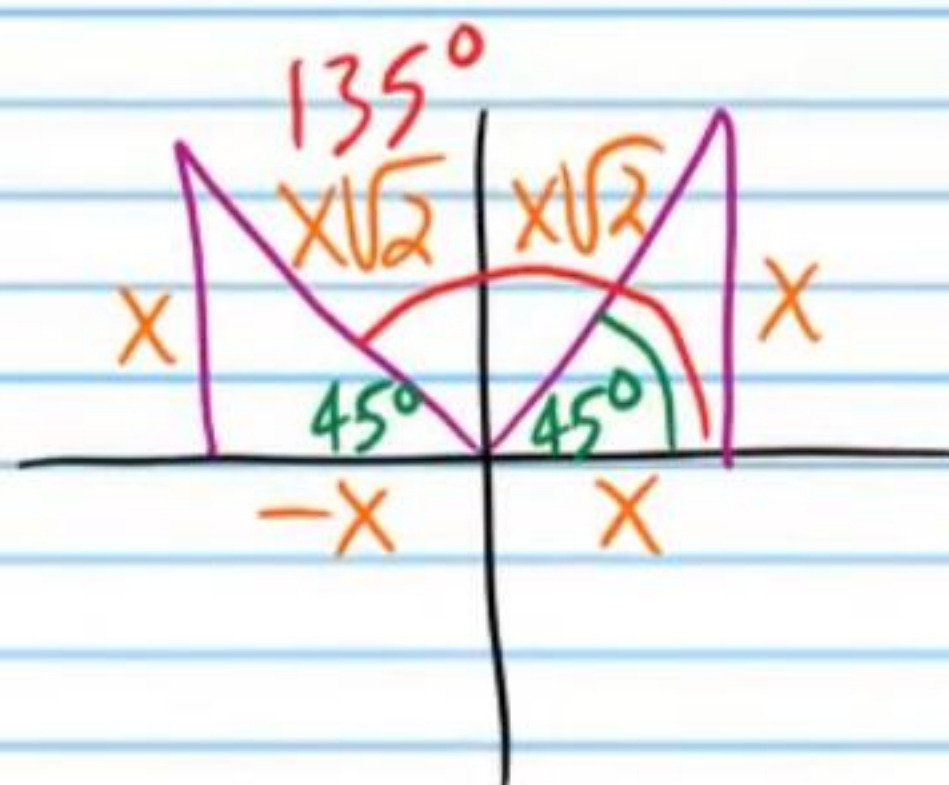
$$\sin^{-1} \frac{1}{\sqrt{2}} = 45^\circ$$

II) $\csc(\sin^{-1} \frac{1}{\sqrt{2}})$

$\csc 45^\circ$

$$\frac{x\sqrt{2}}{x}$$

$$\boxed{\sqrt{2}}$$



⑥ $\sec(\cos^{-1} \frac{\sqrt{3}}{2})$

I) $\theta = \cos^{-1} \frac{\sqrt{3}}{2}$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ, 330^\circ$$

$$\cos^{-1} \frac{\sqrt{3}}{2} = 30^\circ$$

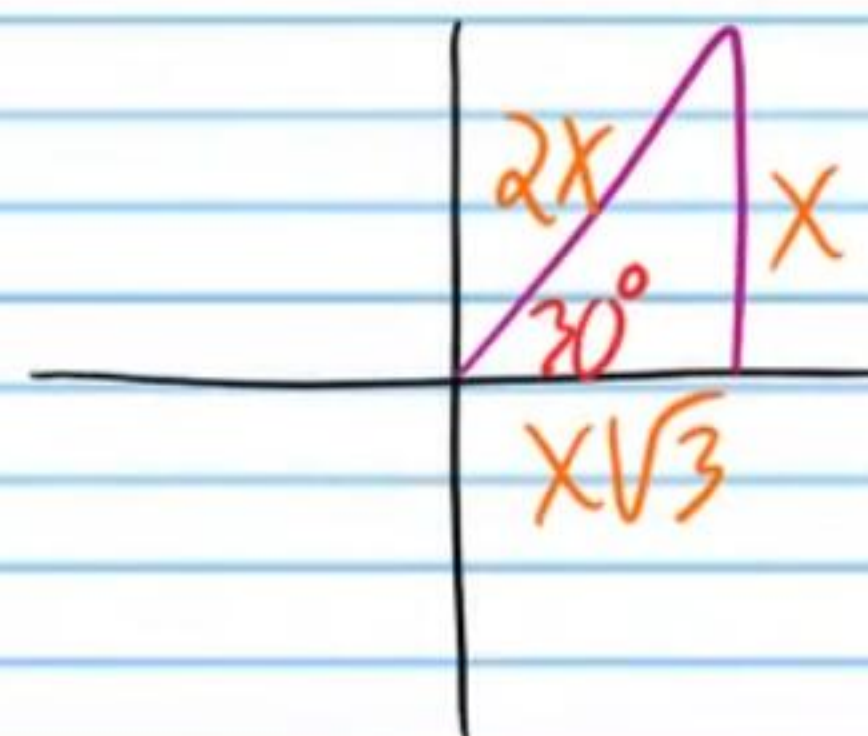
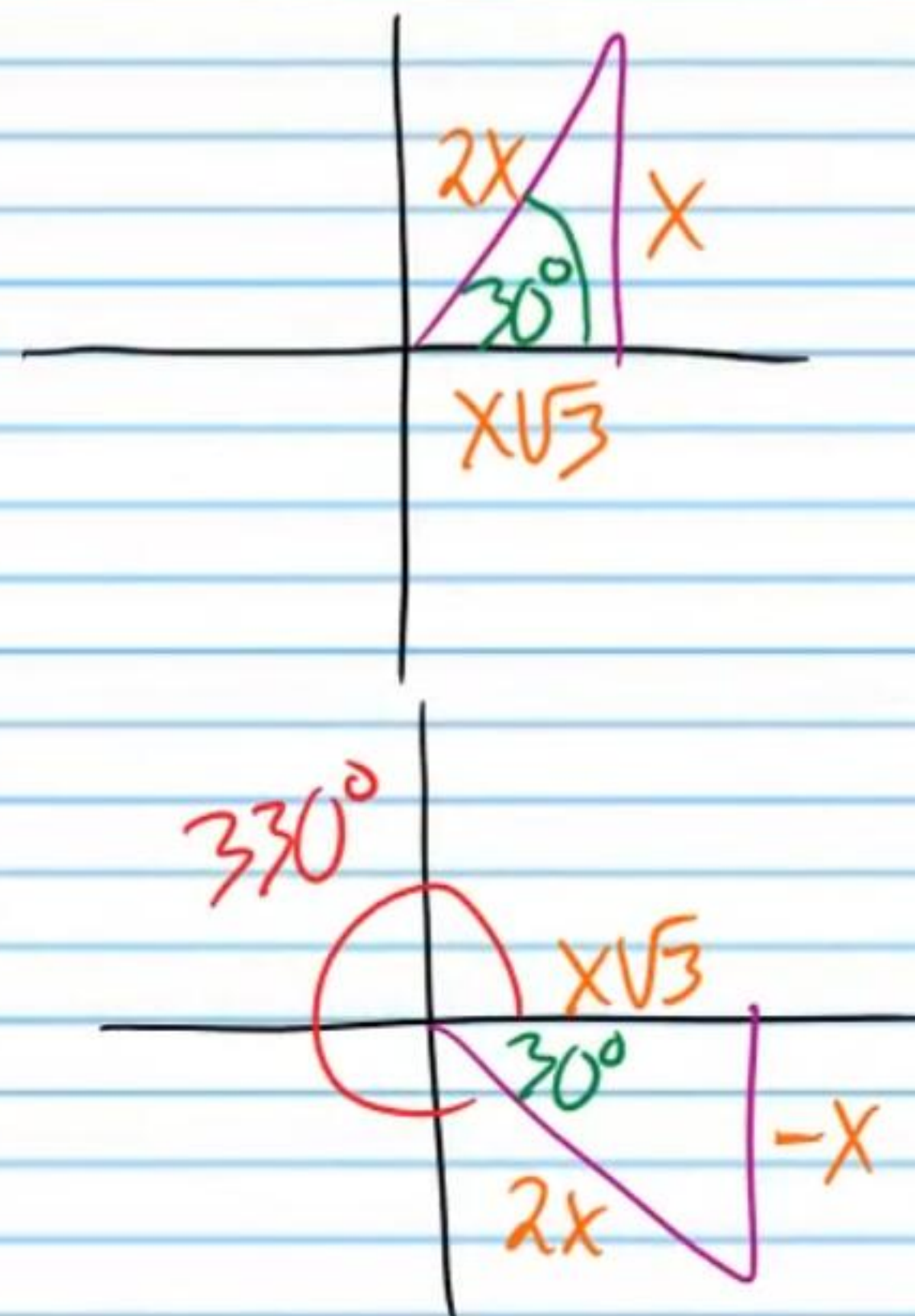
II) $\sec(\cos^{-1} \frac{\sqrt{3}}{2})$

$\sec 30^\circ$

$$\frac{2x}{x\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\boxed{\frac{2\sqrt{3}}{3}}$$



$$\textcircled{7} \sin(\cot^{-1}(0))$$

$$\text{I) } \theta = \cot^{-1}(0)$$

$$\cot \theta = 0$$

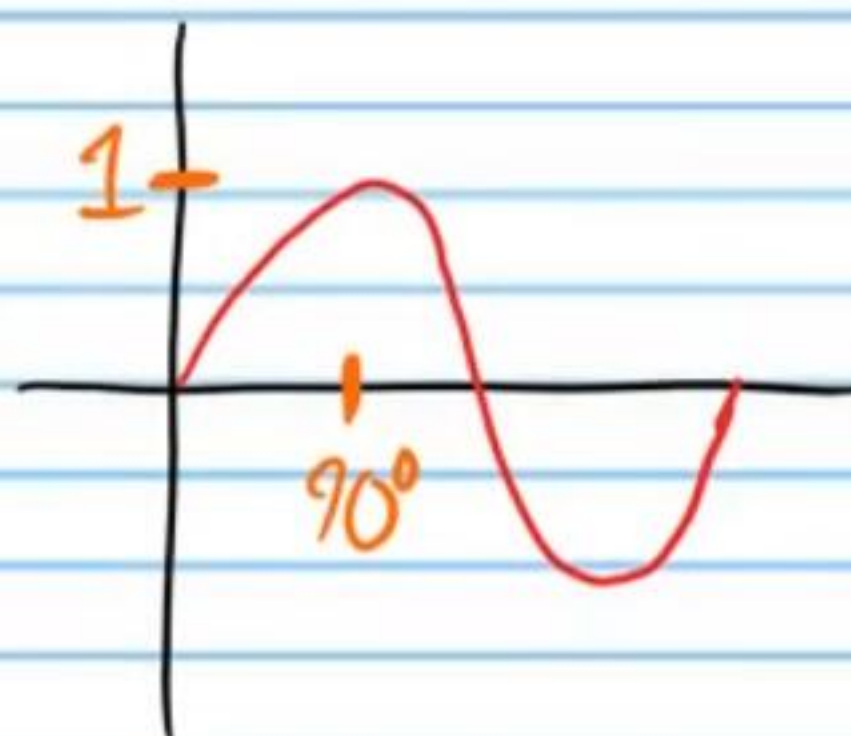
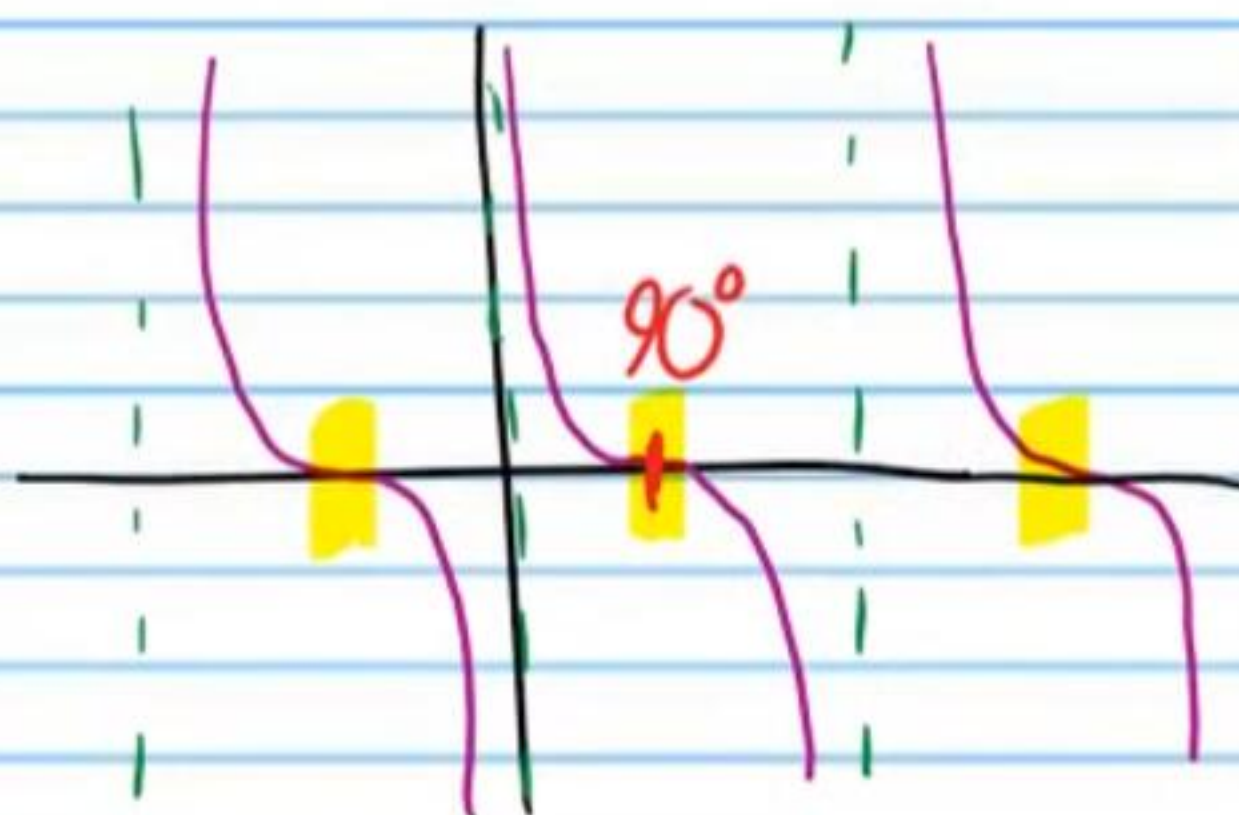
$$\theta = 90^\circ$$

$$\cot^{-1}(0) = 90^\circ$$

$$\text{II) } \sin(\cot^{-1}(0))$$

$$\sin 90^\circ$$

$$\boxed{1}$$



$$\textcircled{8} \cos(\csc^{-1}(\frac{2}{\sqrt{3}}))$$

$$\text{I) } \theta = \csc^{-1}(\frac{2}{\sqrt{3}})$$

$$\csc \theta = \frac{2}{\sqrt{3}}$$

$$\theta = 60^\circ, 120^\circ$$

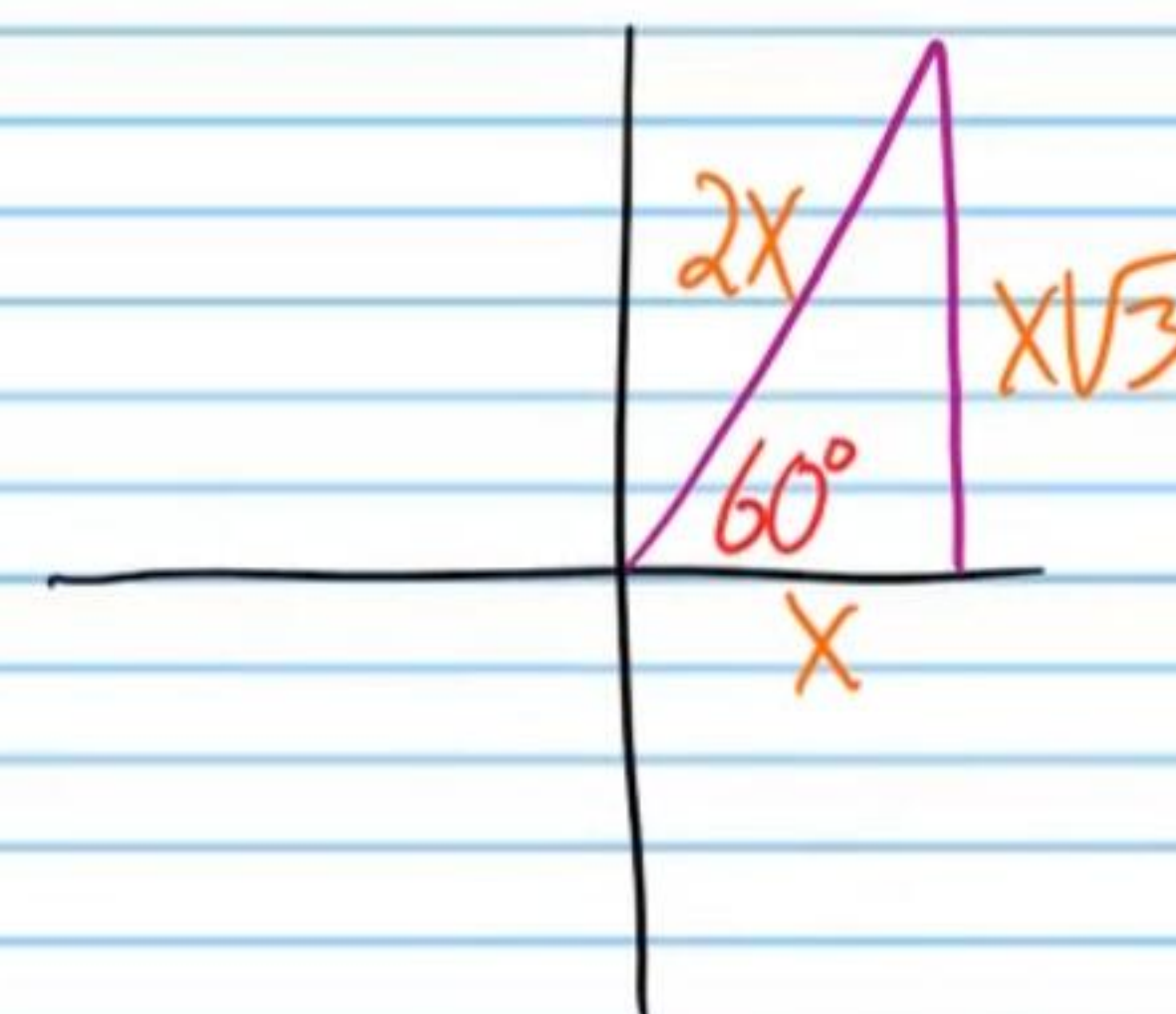
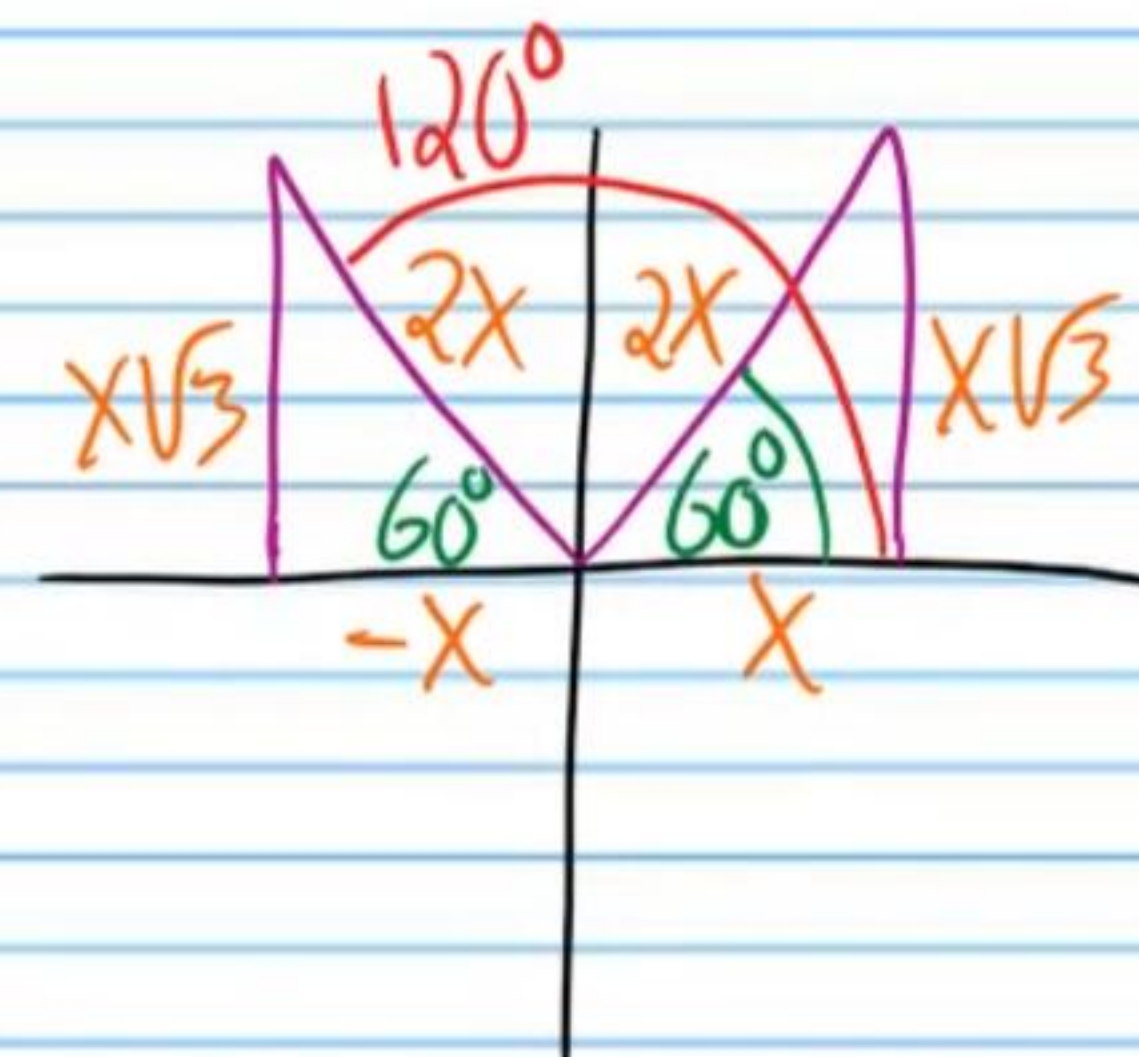
$$\csc^{-1}(\frac{2}{\sqrt{3}}) = 60^\circ$$

$$\text{II) } \cos(\csc^{-1}(\frac{2}{\sqrt{3}}))$$

$$\cos 60^\circ$$

$$\frac{x}{2x}$$

$$\boxed{\frac{1}{2}}$$



The Reference Angle Is Not the Actual Angle

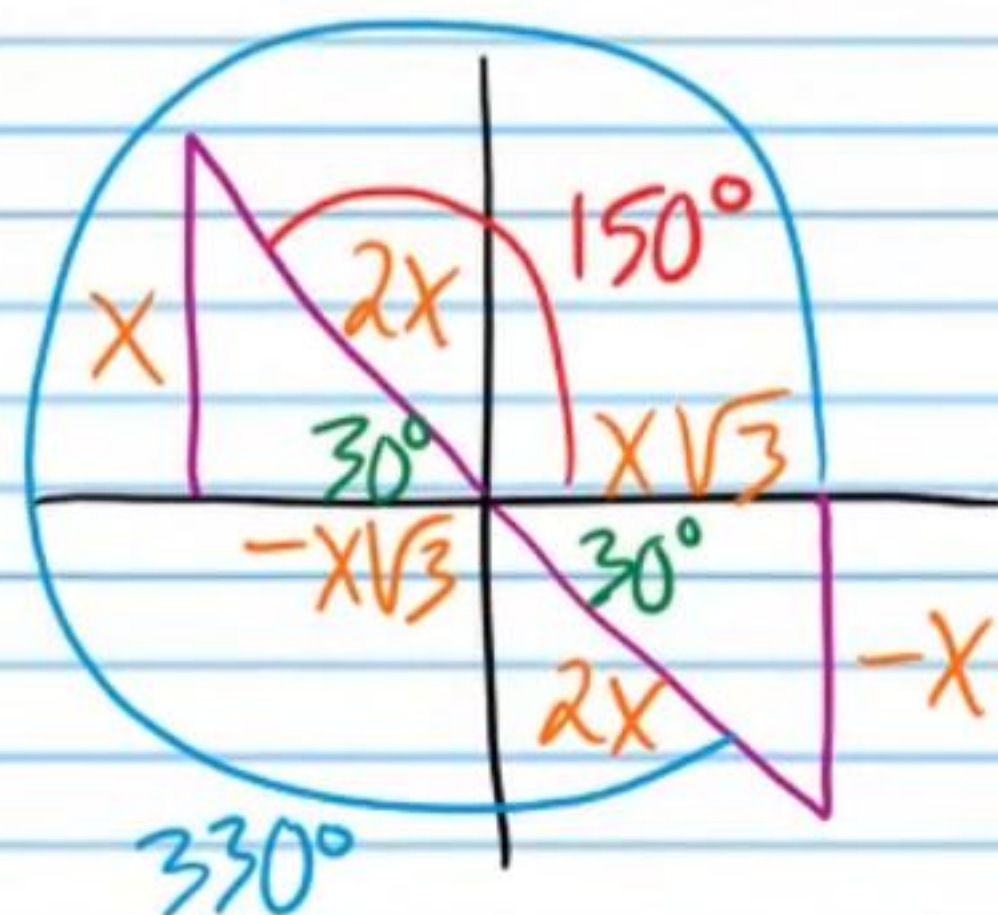
• ⑨ $\csc(\cot^{-1}(-\sqrt{3}))$

I) $\theta = \cot^{-1}(-\sqrt{3})$

$$\cot \theta = -\sqrt{3}$$

$$\theta = 150^\circ, \cancel{330^\circ}$$

$$\cot^{-1}(-\sqrt{3}) = 150^\circ$$

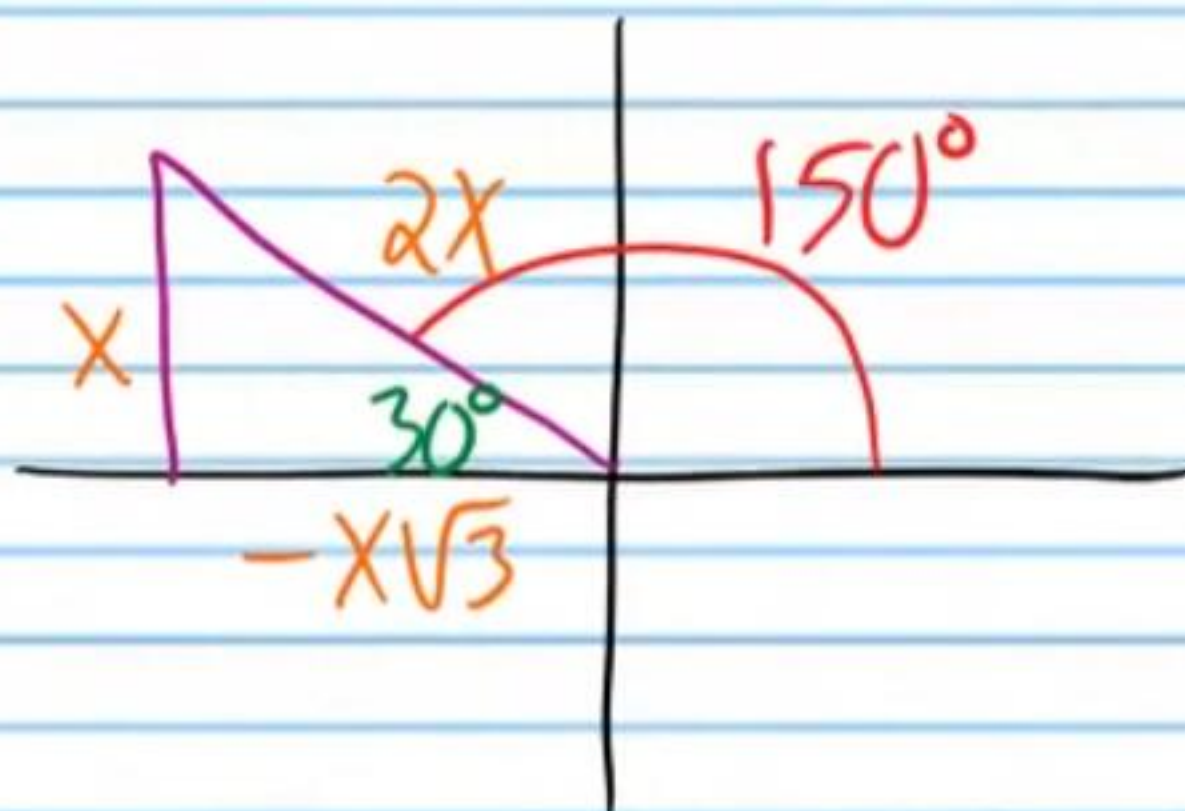


II) $\csc(\cot^{-1}(-\sqrt{3}))$

$$\csc 150^\circ$$

$$\frac{2x}{x}$$

$$\boxed{2}$$



⑩ $\sin(\cos^{-1}(-\frac{1}{\sqrt{2}}))$

I) $\theta = \cos^{-1}(-\frac{1}{\sqrt{2}})$

$$\cos \theta = -\frac{1}{\sqrt{2}}$$

$$\theta = 135^\circ, \cancel{225^\circ}$$

$$\cos^{-1}(-\frac{1}{\sqrt{2}}) = 135^\circ$$

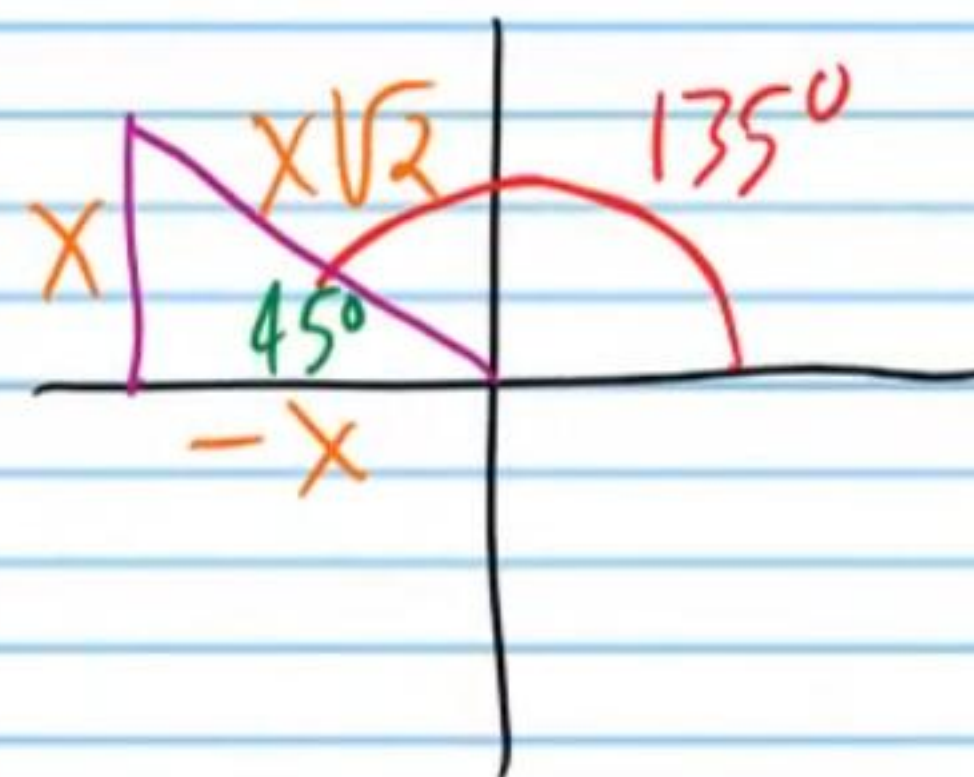
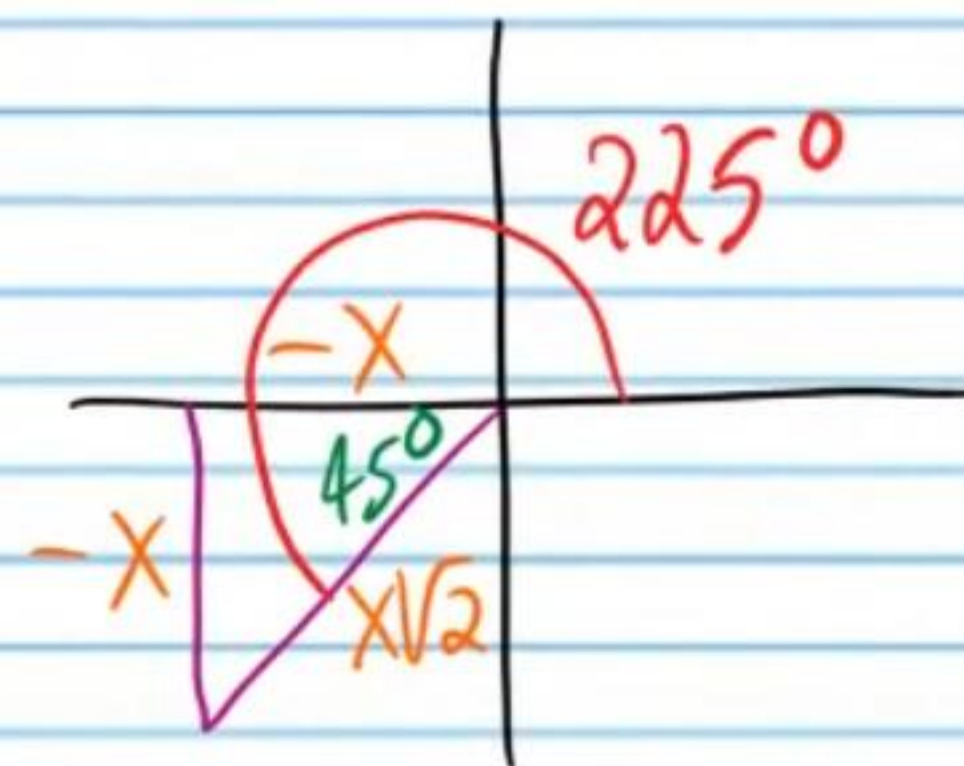
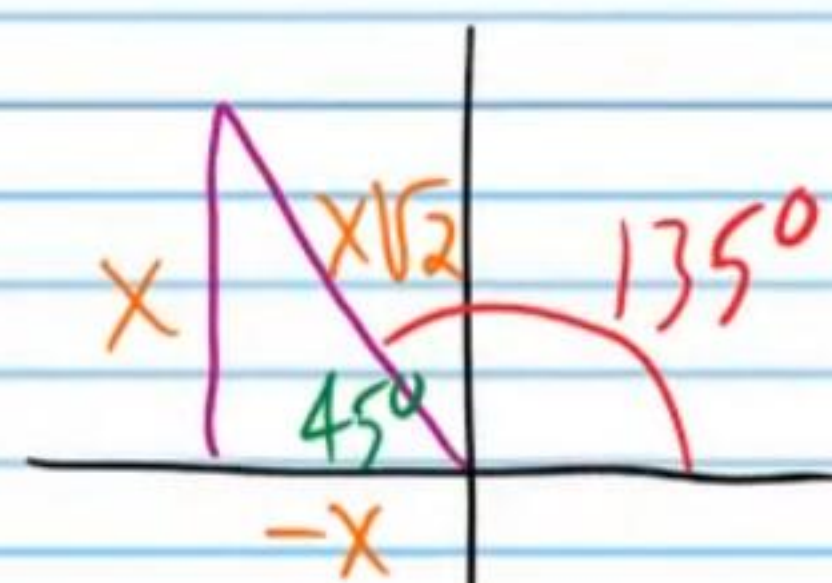
II) $\sin(\cos^{-1}(-\frac{1}{\sqrt{2}}))$

$$\sin 135^\circ$$

$$\frac{x}{x\sqrt{2}}$$

$$\frac{1}{\sqrt{2}}$$

$$\boxed{\frac{\sqrt{2}}{2}}$$



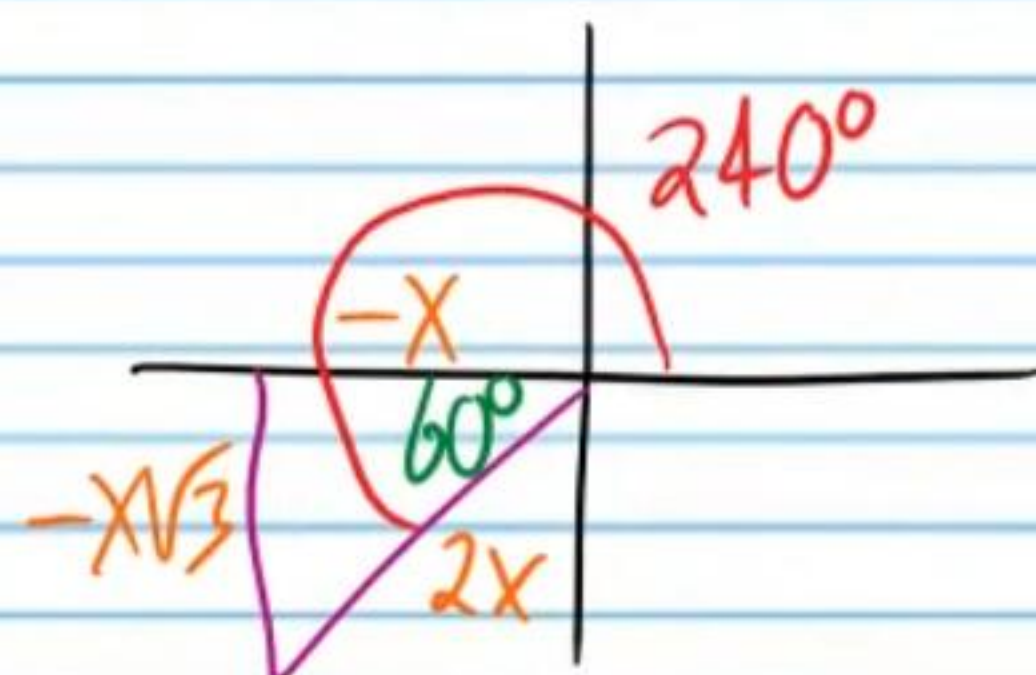
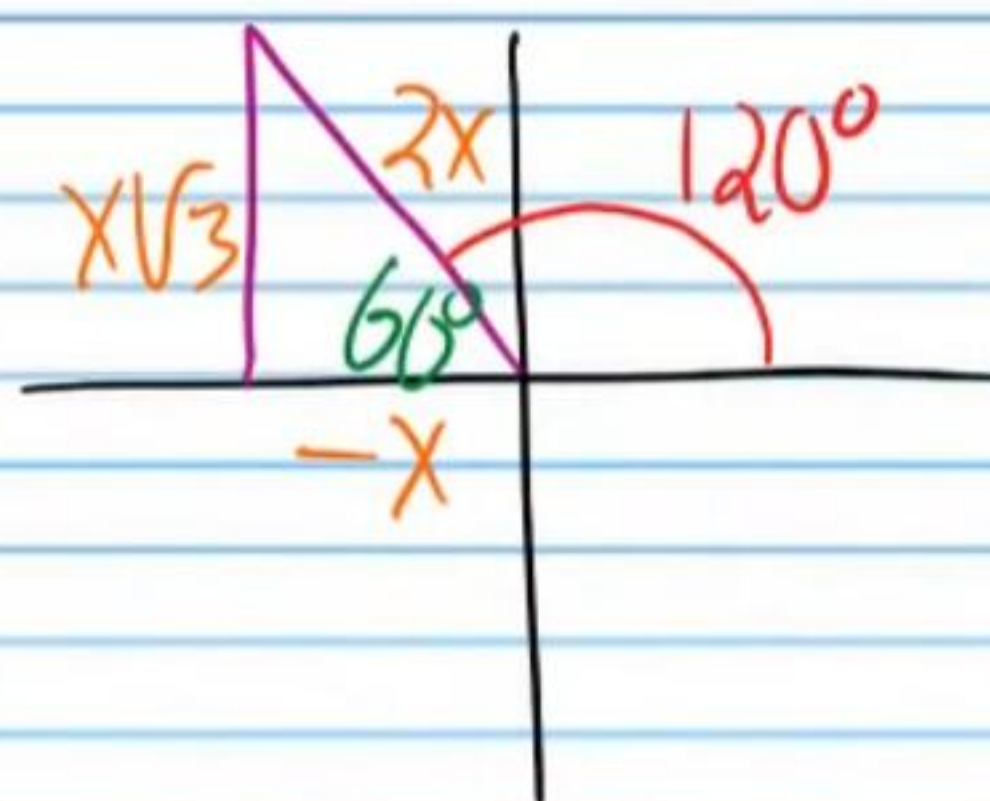
• ⑪ $\tan(\sec^{-1}(-2))$

I) $\theta = \sec^{-1}(-2)$

$\sec \theta = -2$

$\theta = 120^\circ, 240^\circ$

$\sec^{-1}(-2) = 120^\circ$

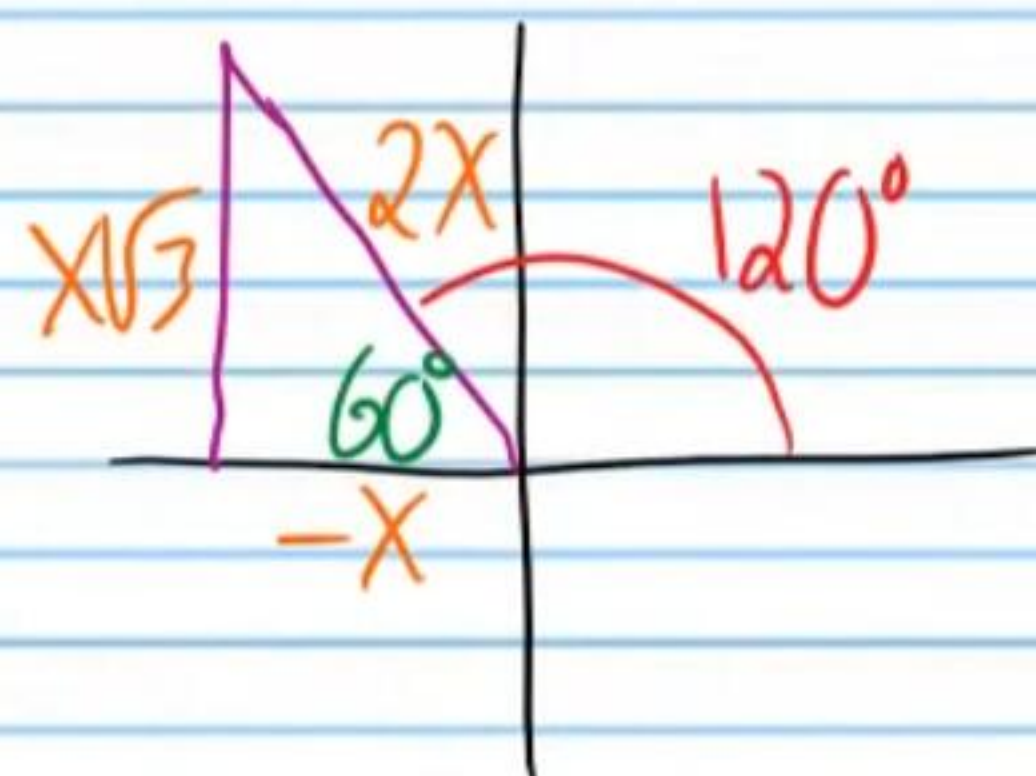


II) $\tan(\sec^{-1}(-2))$

$\tan 120^\circ$

$\frac{x\sqrt{3}}{-x}$

$\boxed{-\sqrt{3}}$



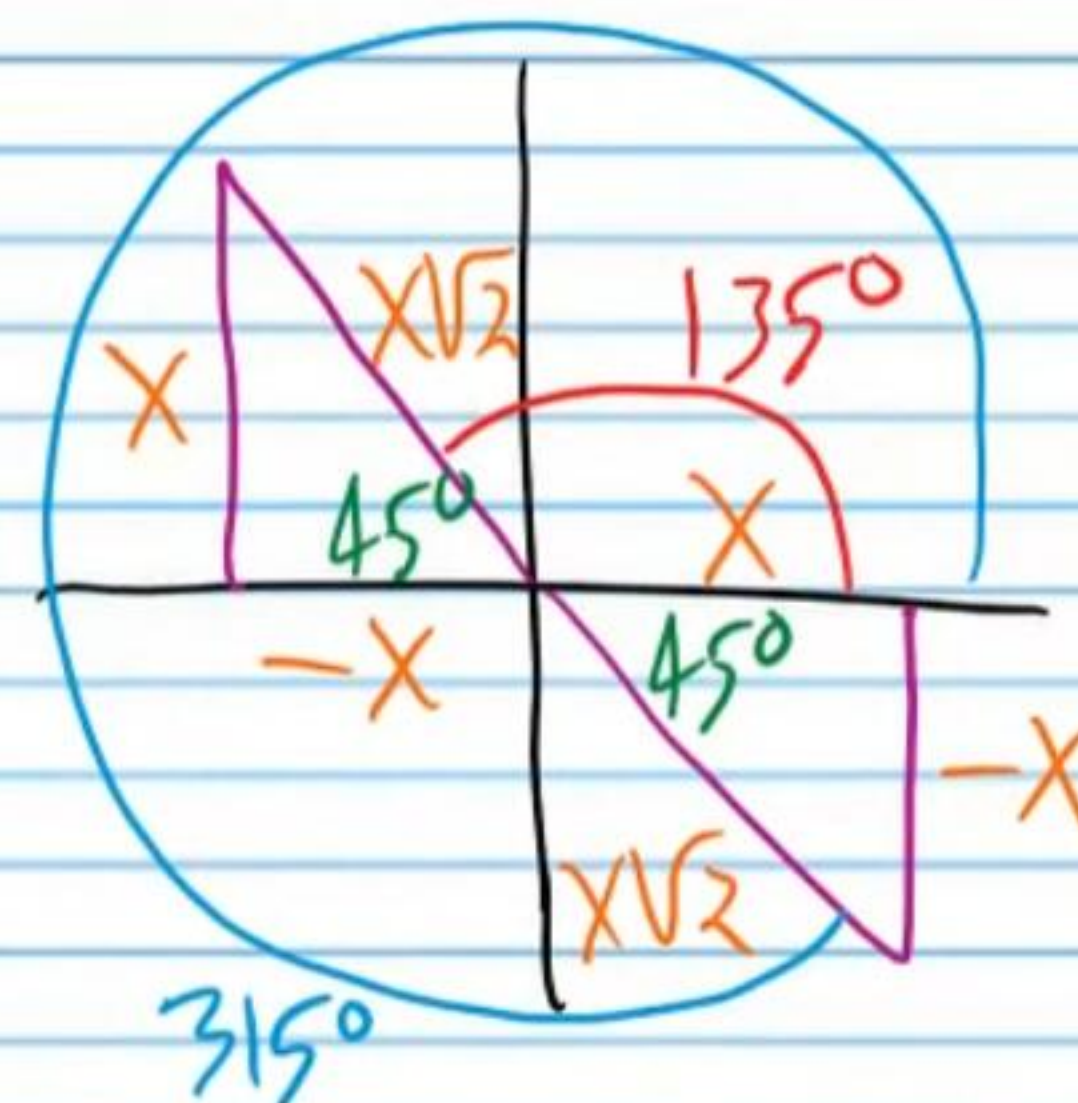
⑫ $\cos(\cot^{-1}(-1))$

I) $\theta = \cot^{-1}(-1)$

$\cot \theta = -1$

$\theta = 135^\circ, 315^\circ$

$\cot^{-1}(-1) = 135^\circ$



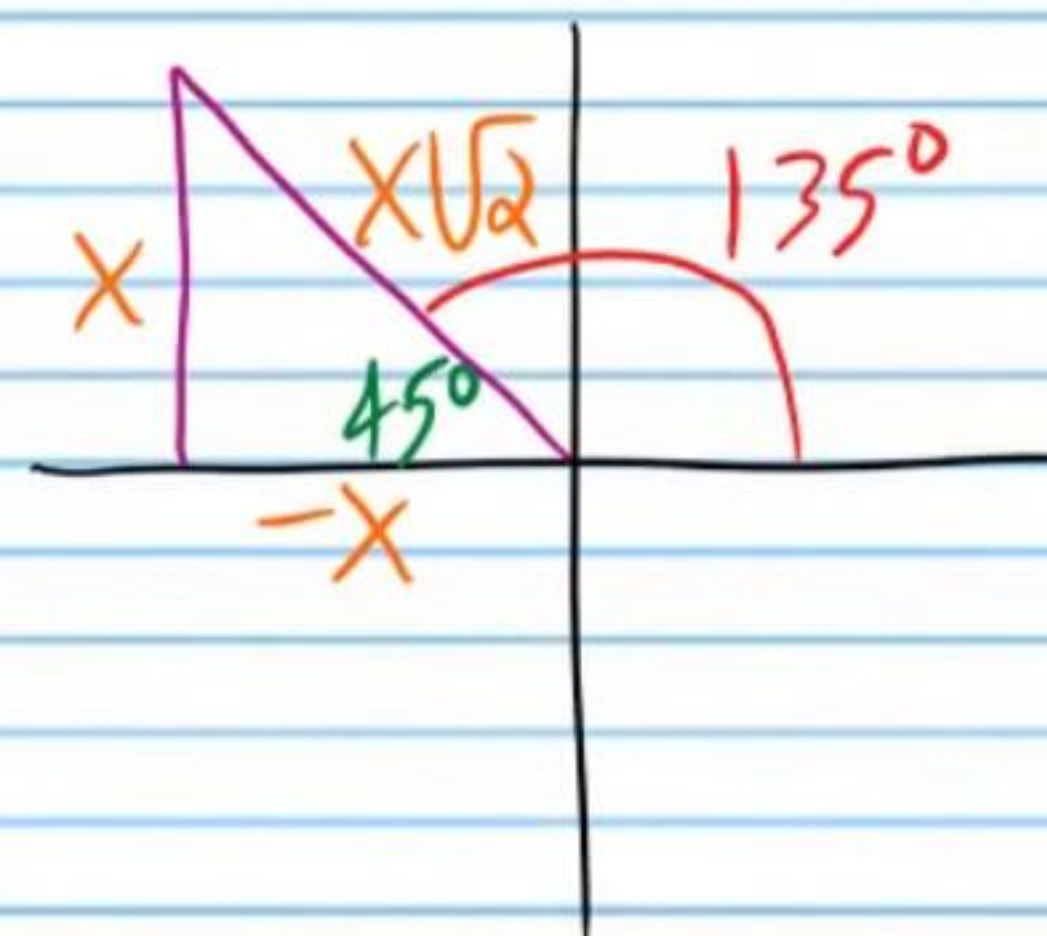
II) $\cos(\cot^{-1}(-1))$

$\cos 135^\circ$

$\frac{-1}{\sqrt{2}}$

$-\frac{1}{\sqrt{2}}$

$\boxed{-\frac{\sqrt{2}}{2}}$



The Inner Expressions Will Be **Negative** Angle Measures

• ⑬ $\cot(\csc^{-1}(-\sqrt{2}))$

I) $\theta = \csc^{-1}(-\sqrt{2})$

$$\csc \theta = -\sqrt{2}$$

$$\theta = \cancel{225^\circ}, \cancel{315^\circ}$$

$$\theta = -45^\circ$$

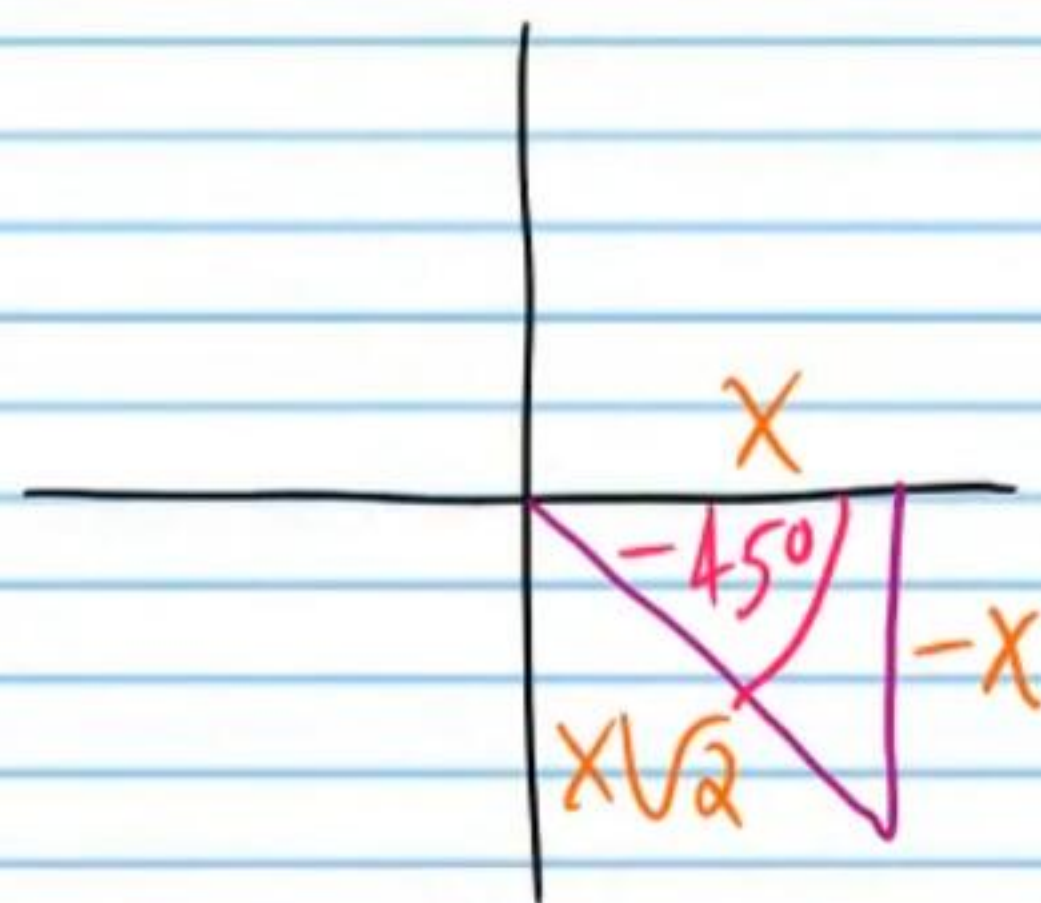
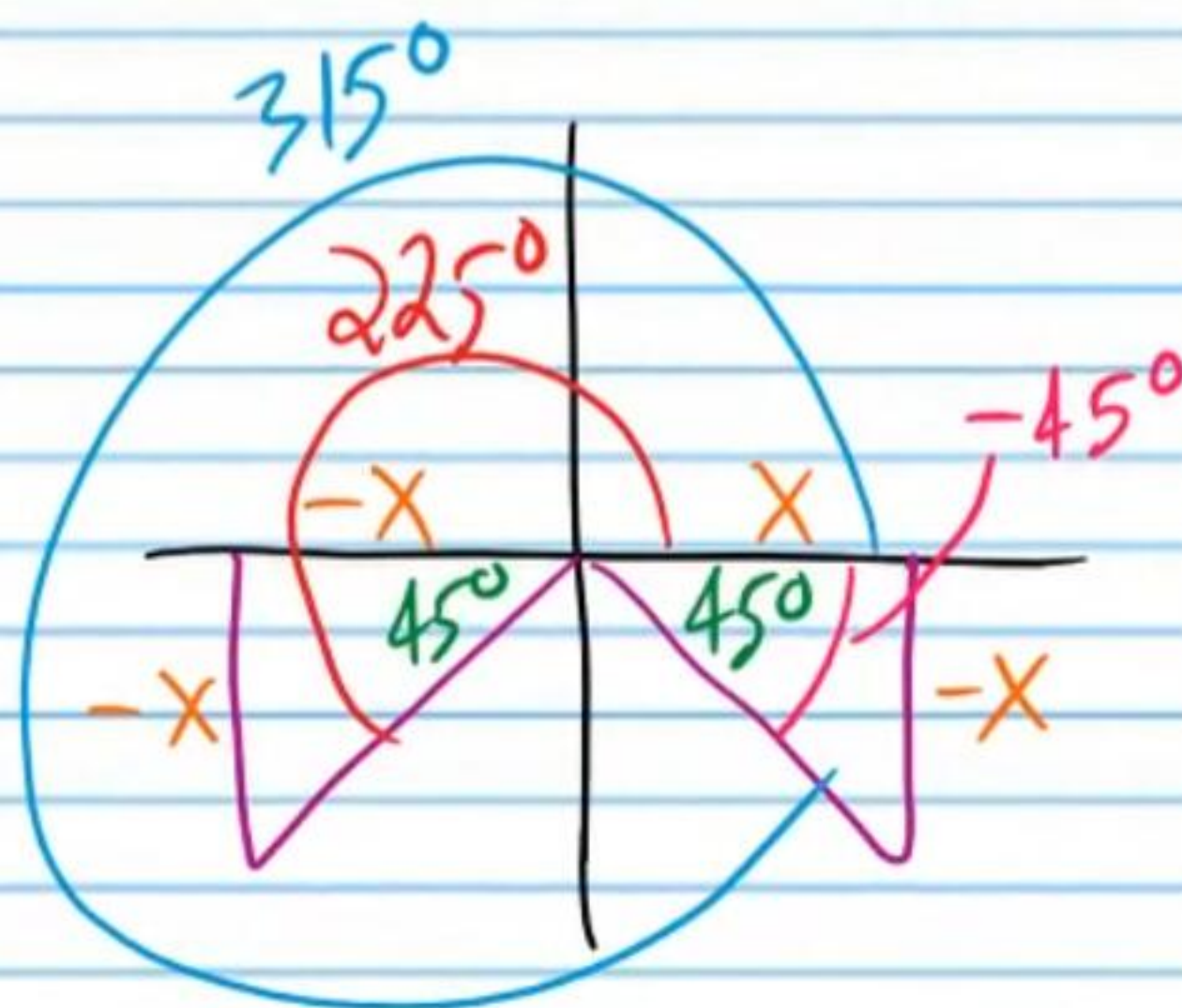
$$\csc^{-1}(-\sqrt{2}) = -45^\circ$$

II) $\cot(\csc^{-1}(-\sqrt{2}))$

$$\cot(-45^\circ)$$

$$\frac{x}{-x}$$

$$\boxed{-1}$$



⑭ $\cos(\sin^{-1}(-\frac{\sqrt{3}}{2}))$

I) $\theta = \sin^{-1}(-\frac{\sqrt{3}}{2})$

$$\sin \theta = -\frac{\sqrt{3}}{2}$$

$$\theta = \cancel{240^\circ}, \cancel{300^\circ}$$

$$\theta = -60^\circ$$

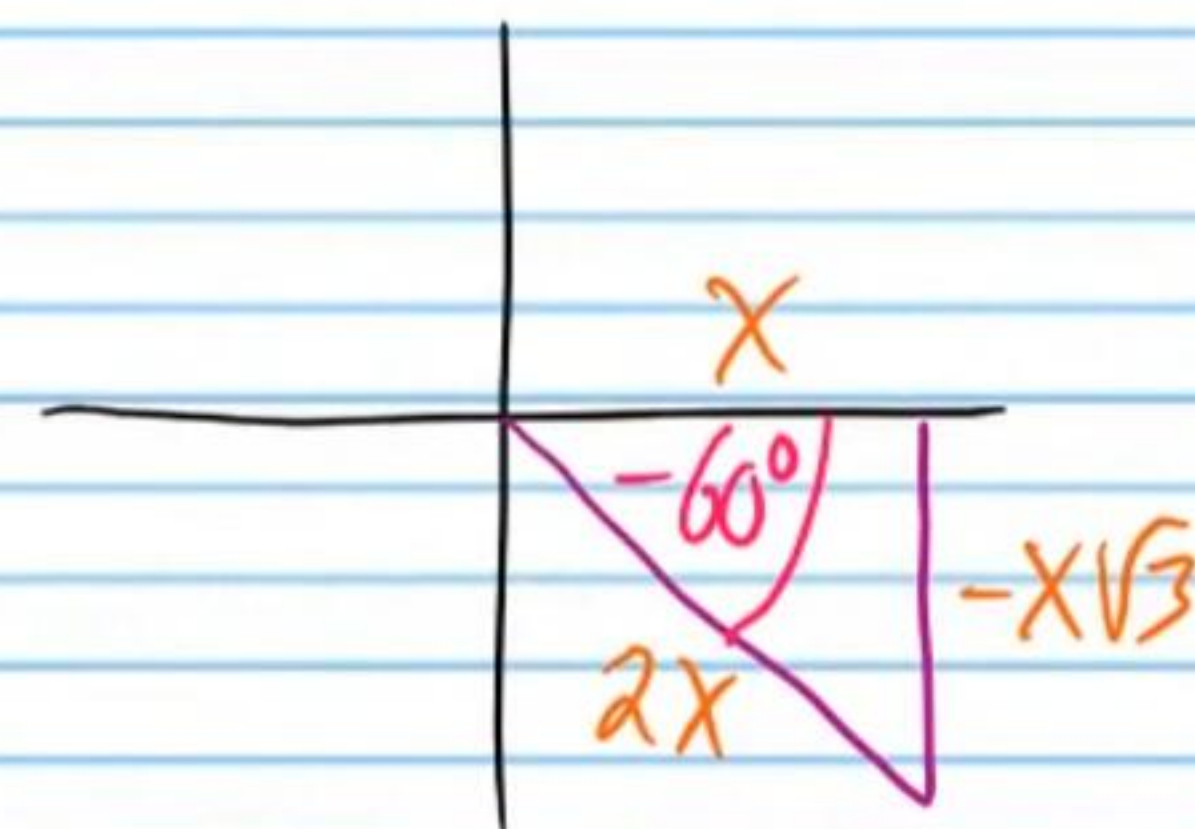
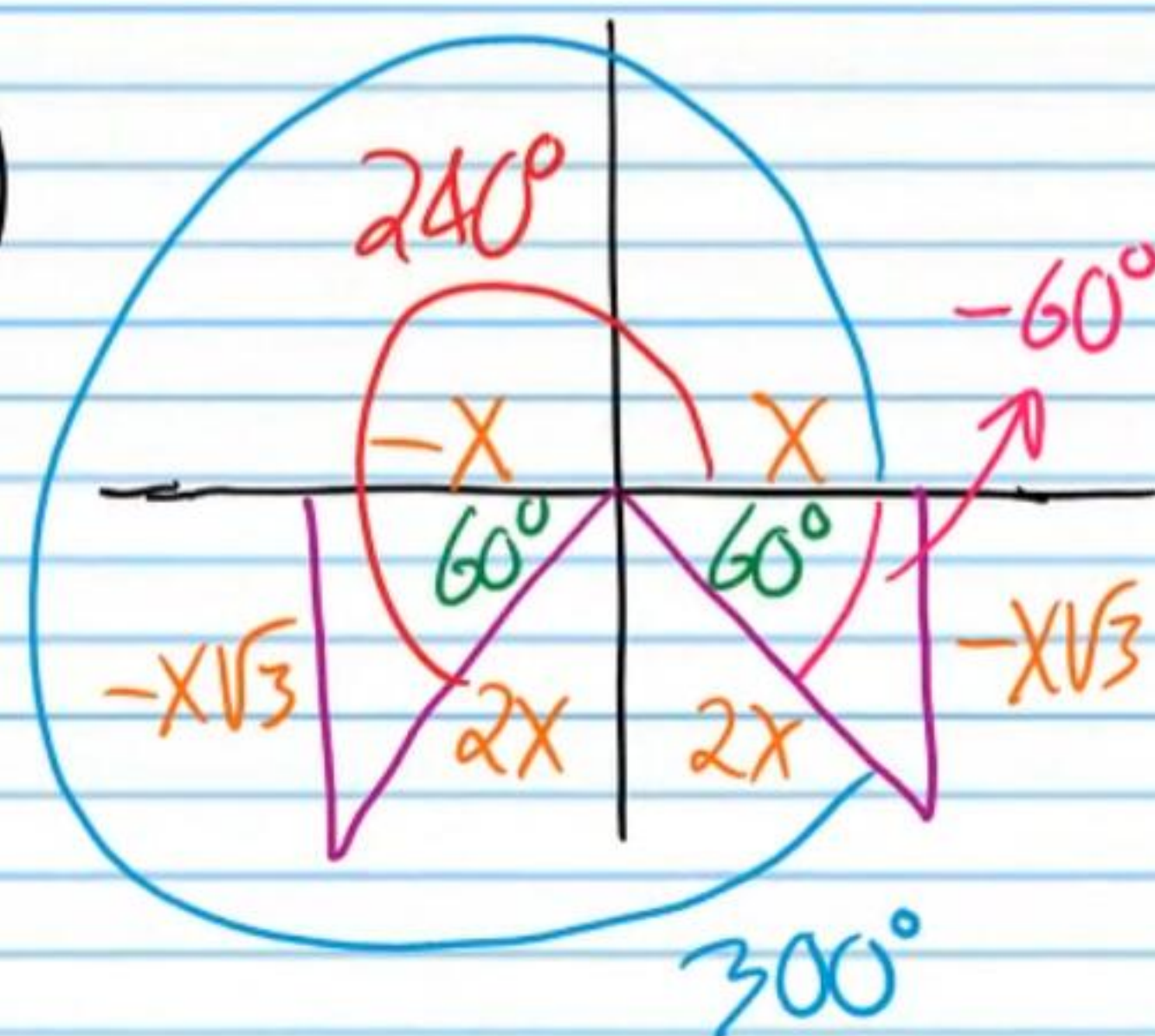
$$\sin^{-1}(-\frac{\sqrt{3}}{2}) = -60^\circ$$

II) $\cos(\sin^{-1}(-\frac{\sqrt{3}}{2}))$

$$\cos(-60^\circ)$$

$$\frac{x}{2x}$$

$$\boxed{\frac{1}{2}}$$



• ⑮ $\cos(\tan^{-1}(-\frac{1}{\sqrt{3}}))$

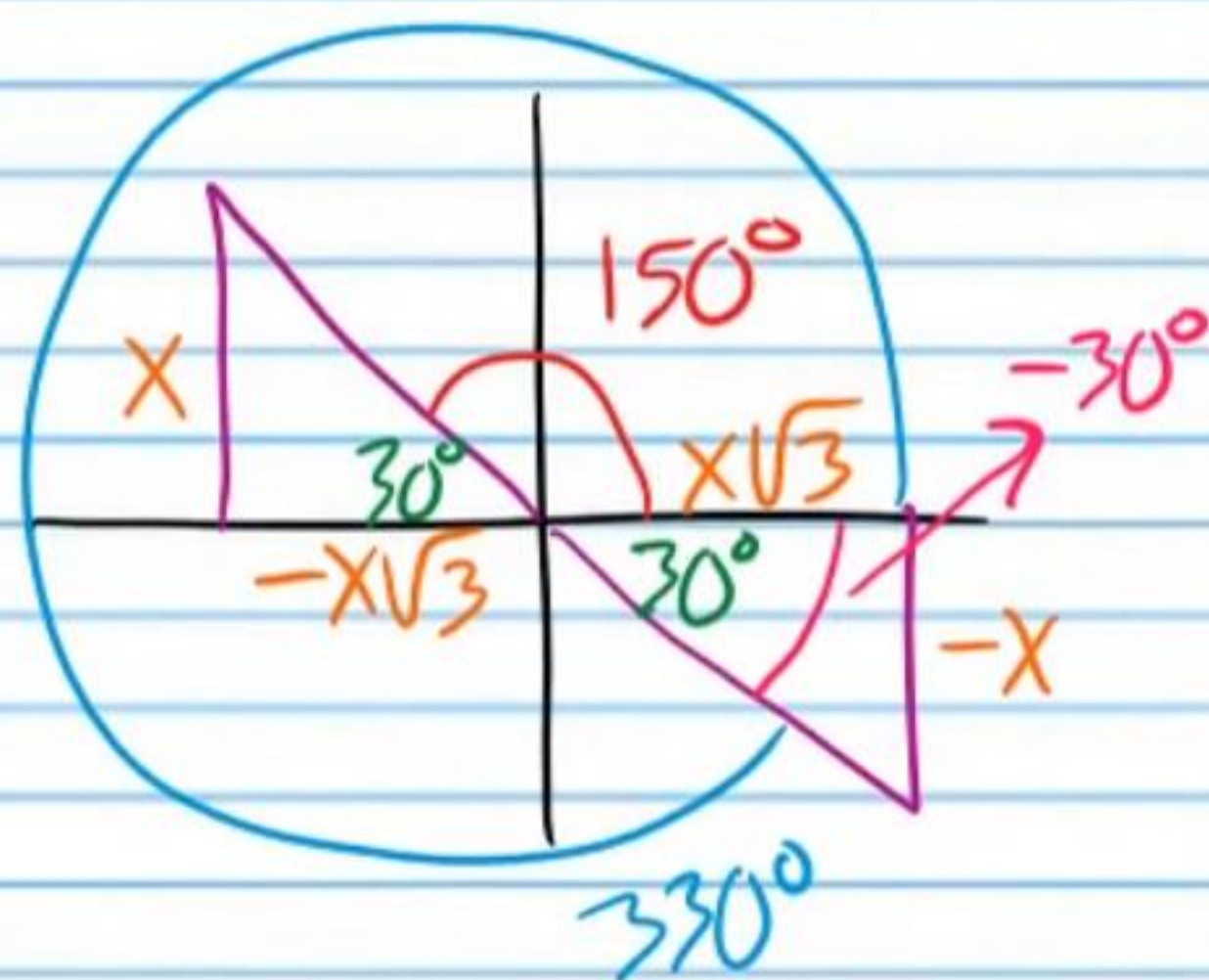
I) $\theta = \tan^{-1}(-\frac{1}{\sqrt{3}})$

$$\tan \theta = -\frac{1}{\sqrt{3}}$$

$$\theta = 150^\circ, 330^\circ$$

$$\theta = -30^\circ$$

$$\tan^{-1}(-\frac{1}{\sqrt{3}}) = -30^\circ$$

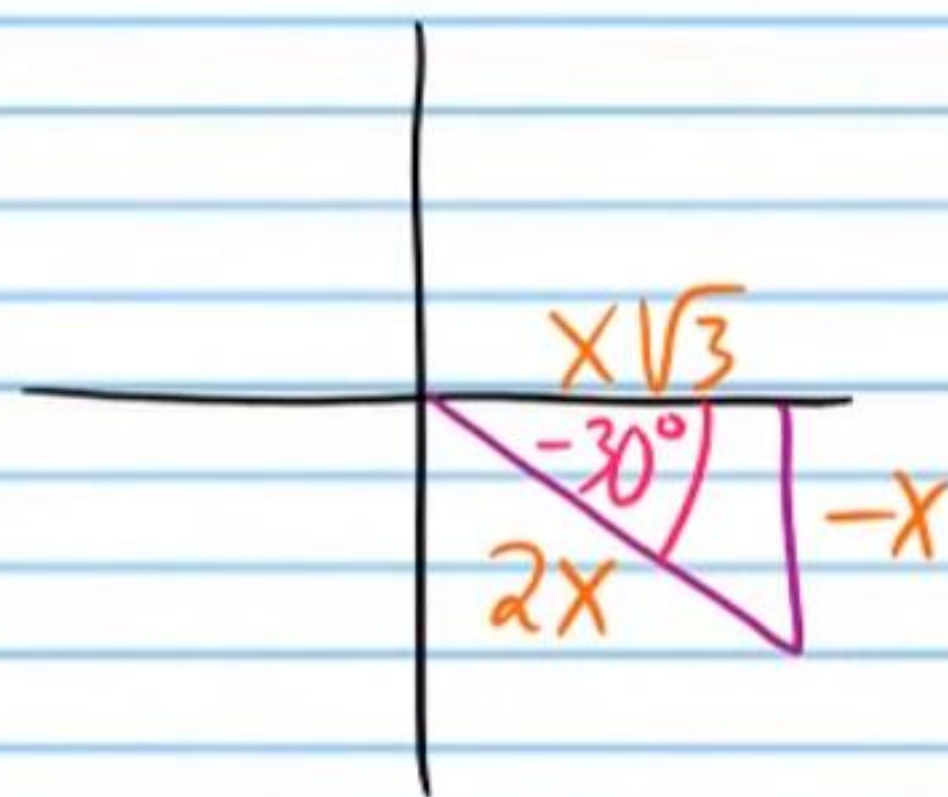


II) $\cos(\tan^{-1}(-\frac{1}{\sqrt{3}}))$

$$\cos(-30^\circ)$$

$$\frac{X\sqrt{3}}{2X}$$

$$\boxed{\frac{\sqrt{3}}{2}}$$



⑯ $\csc(\sin^{-1}(-\frac{1}{\sqrt{2}}))$

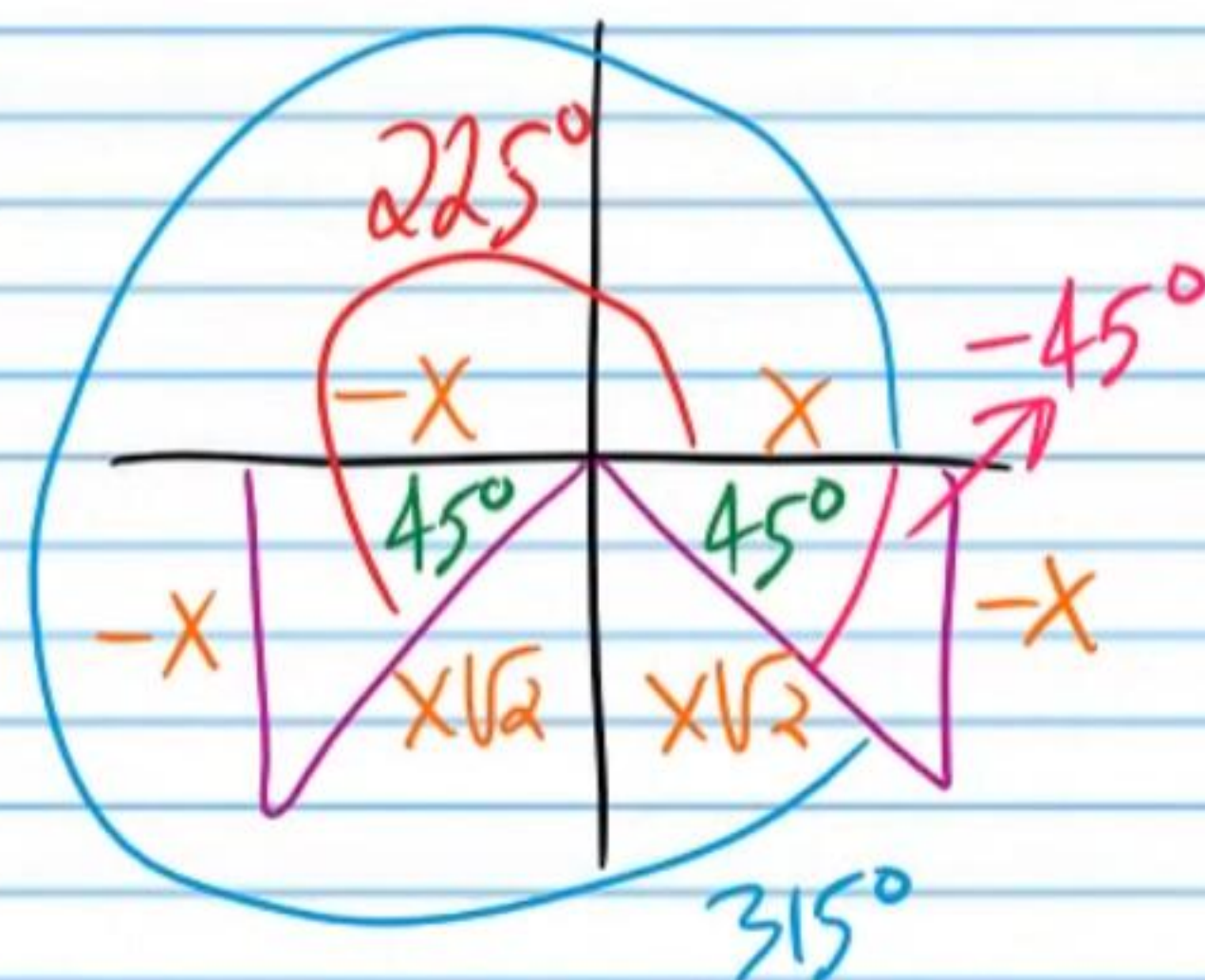
I) $\theta = \sin^{-1}(-\frac{1}{\sqrt{2}})$

$$\sin \theta = -\frac{1}{\sqrt{2}}$$

$$\theta = 225^\circ, 315^\circ$$

$$\theta = -45^\circ$$

$$\sin^{-1}(-\frac{1}{\sqrt{2}}) = -45^\circ$$

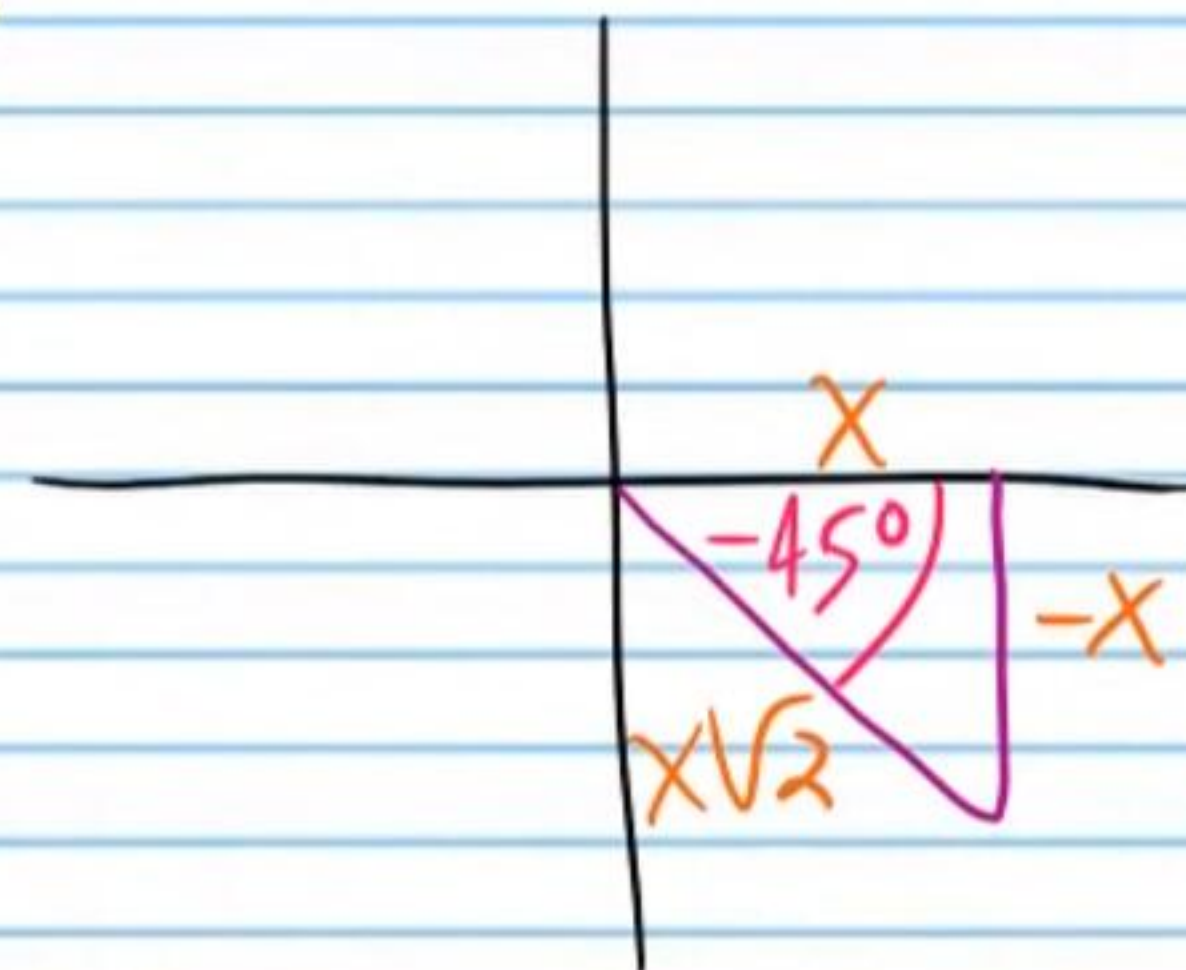


II) $\csc(\sin^{-1}(-\frac{1}{\sqrt{2}}))$

$$\csc(-45^\circ)$$

$$\frac{X\sqrt{2}}{-X}$$

$$\boxed{-\sqrt{2}}$$



• (17) $\cos(\cos^{-1}(-\frac{1}{2}))$

I) $\theta = \cos^{-1}(-\frac{1}{2})$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = 120^\circ, 240^\circ$$

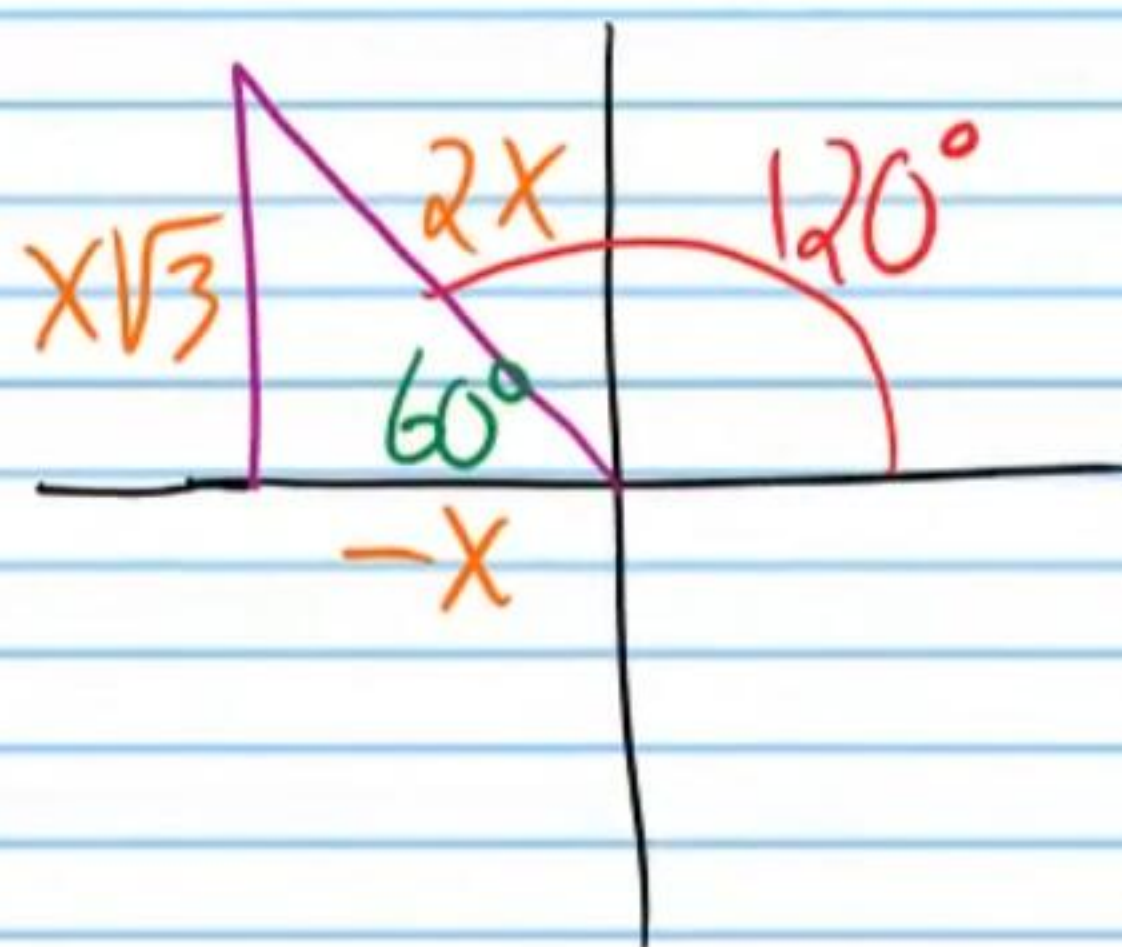
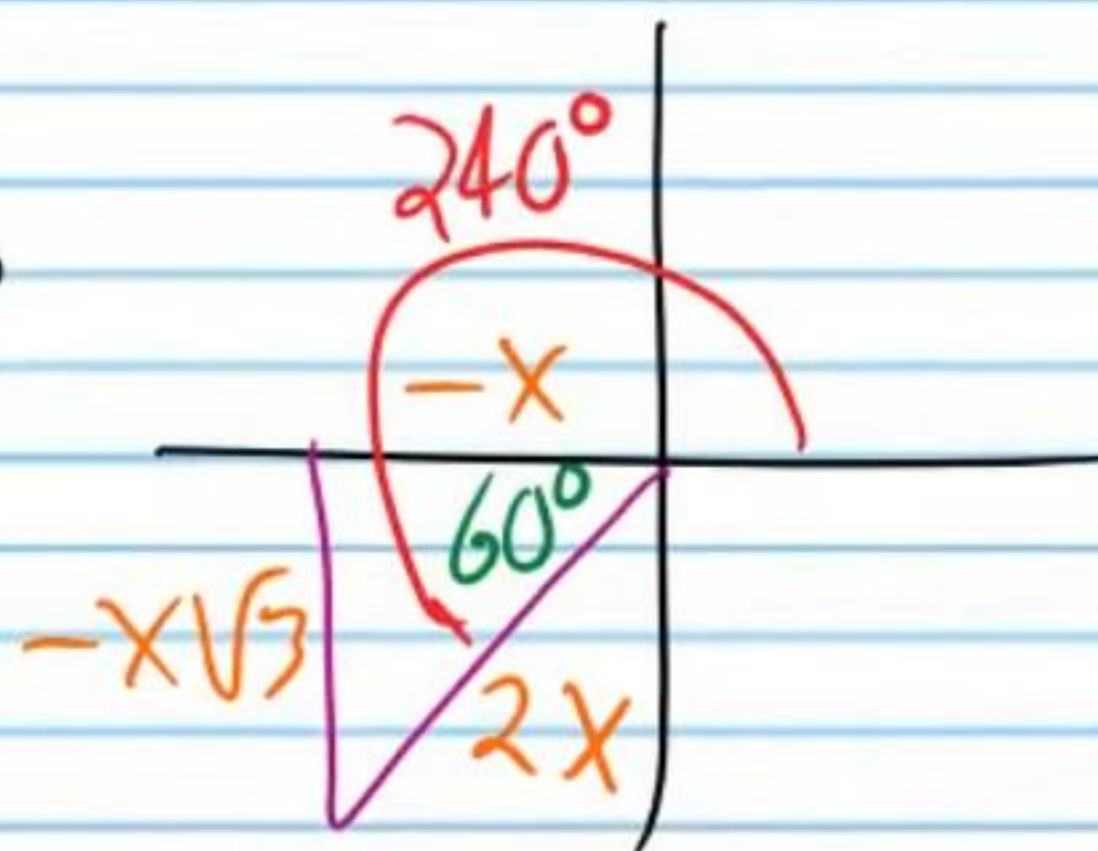
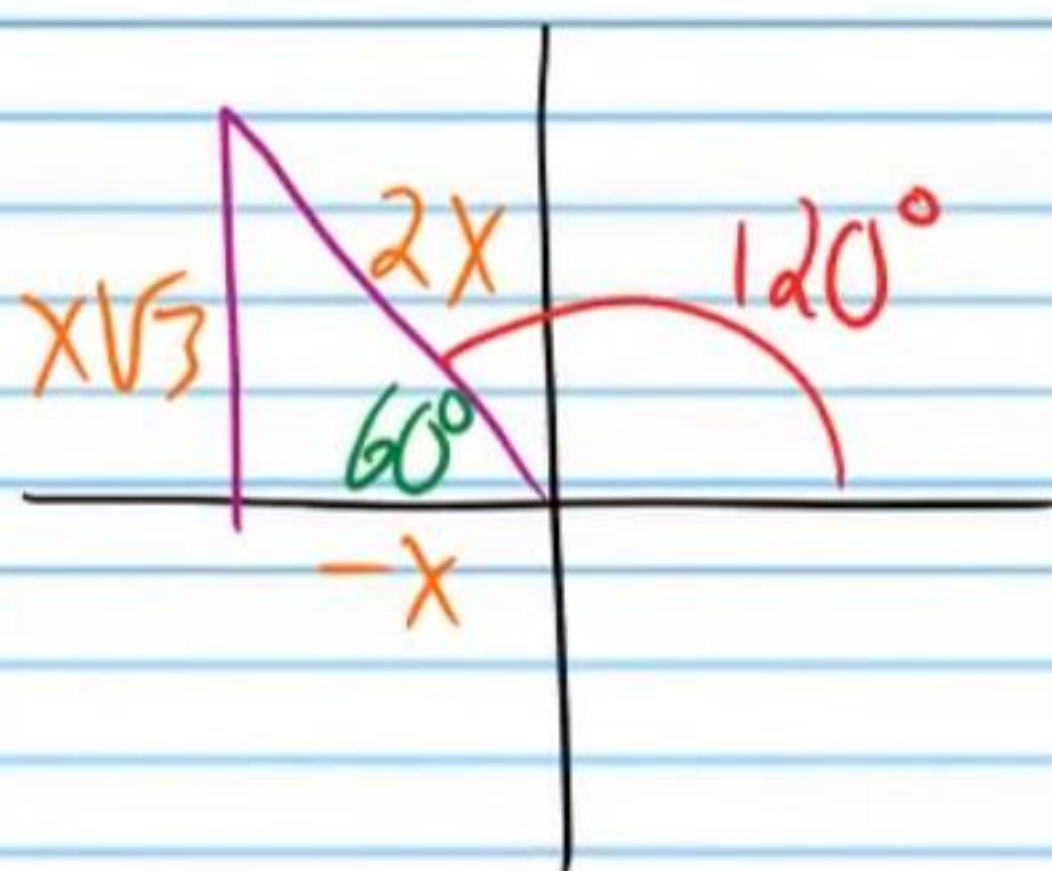
$$\cos^{-1}(-\frac{1}{2}) = 120^\circ$$

II) $\cos(\cos^{-1}(-\frac{1}{2}))$

$$\cos 120^\circ$$

$$\frac{-x}{2x}$$

$$\boxed{-\frac{1}{2}}$$



(18) $\cot(\cot^{-1}(1))$

$$\boxed{1}$$

(19) $\csc(\csc^{-1}(1))$

$$\boxed{1}$$

• (20) $\sin(\sin^{-1}(-\frac{1}{2}))$

$$\boxed{-\frac{1}{2}}$$

(21) $\tan(\tan^{-1}(0))$

$$\boxed{0}$$

(22) $\sec(\sec^{-1}\sqrt{2})$

$$\boxed{\sqrt{2}}$$

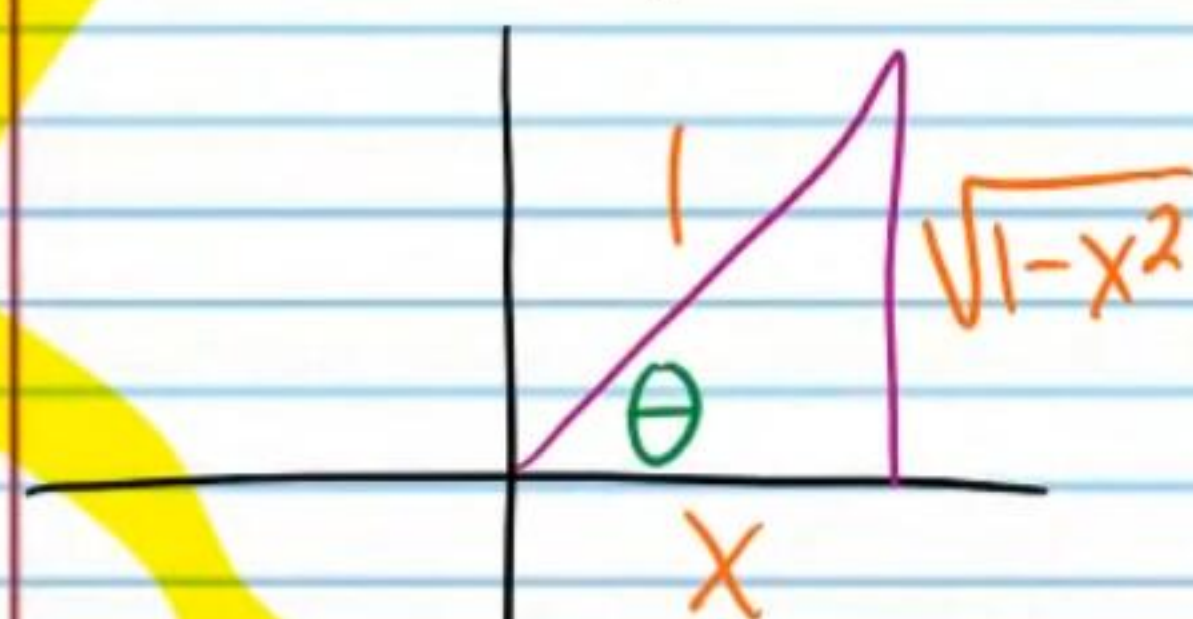
Changing the Form of Composite Trigonometric Expressions

Write an equivalent expression for each problem without trigonometric notation. Assume that x is positive.

• ②③ $\sin(\sec^{-1} \frac{1}{x})$

I) $\theta = \sec^{-1} \frac{1}{x}$

$$\sec \theta = \frac{1}{x}$$



$$a^2 + b^2 = c^2$$

$$x^2 + b^2 = 1^2$$

$$x^2 + b^2 = 1$$

$$b^2 = 1 - x^2$$

$$b = \pm \sqrt{1 - x^2}$$

$$b = \sqrt{1 - x^2}$$

II) $\sin(\sec^{-1} \frac{1}{x})$

$$\sin \theta$$

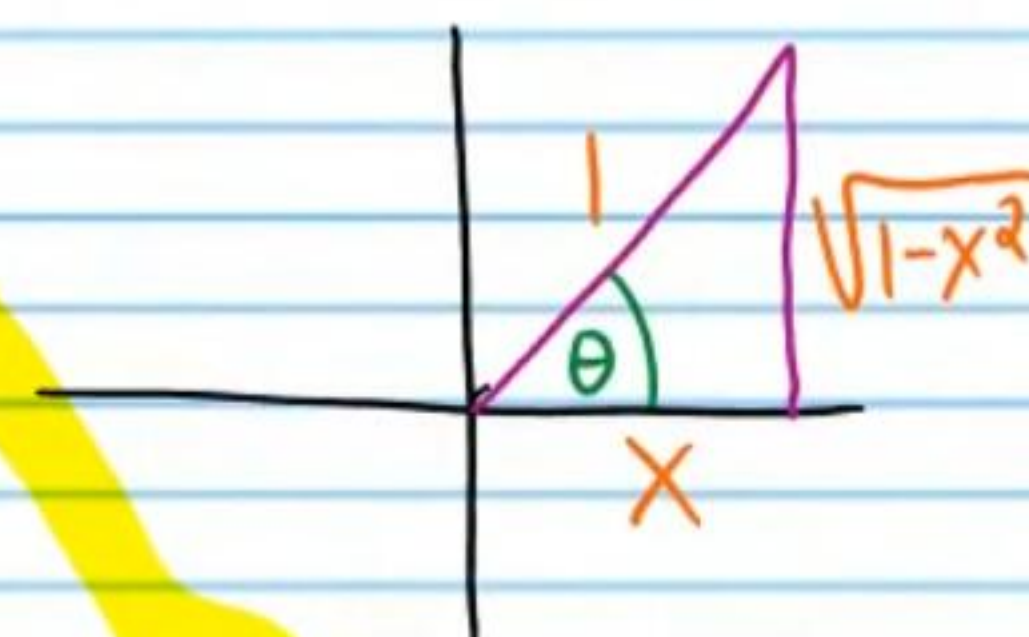
$$\frac{\sqrt{1-x^2}}{1}$$

$$\boxed{\sqrt{1-x^2}}$$

②④ $\csc(\cos^{-1} x)$

I) $\theta = \cos^{-1} x$

$$\cos \theta = x = \frac{x}{1}$$



$$a^2 + b^2 = c^2$$

$$x^2 + b^2 = 1^2$$

$$x^2 + b^2 = 1$$

$$b^2 = 1 - x^2$$

$$b = \pm \sqrt{1 - x^2}$$

$$b = \sqrt{1 - x^2}$$

II) $\csc(\cos^{-1} x)$

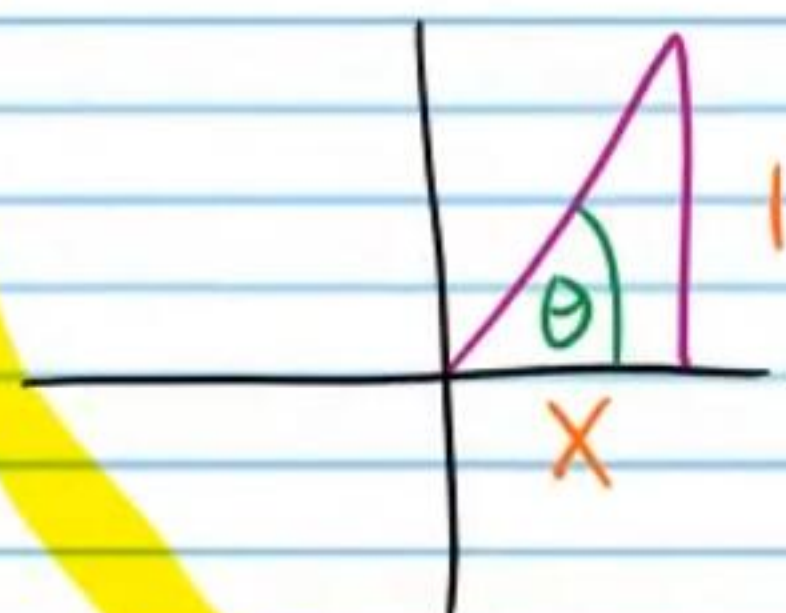
$$\csc \theta$$

$$\boxed{\frac{1}{\sqrt{1-x^2}}}$$

②⑤ $\cot(\tan^{-1} \frac{1}{x})$

I) $\theta = \tan^{-1} \frac{1}{x}$

$\tan \theta = \frac{1}{x}$



II) $\cot(\tan^{-1} \frac{1}{x})$

$\cot \theta$

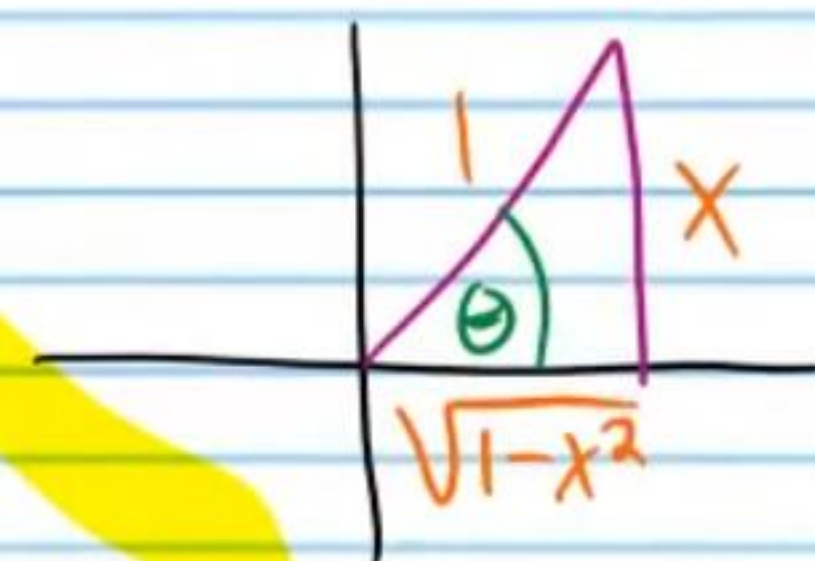
$\frac{x}{1}$

$\boxed{x}$

• ②⑥ $\tan(\sin^{-1} x)$

I) $\theta = \sin^{-1} x$

$\sin \theta = x = \frac{x}{1}$



II) $\tan(\sin^{-1} x)$

$\tan \theta$

$\boxed{\frac{x}{\sqrt{1-x^2}}}$

$a^2 + b^2 = c^2$

$x^2 + b^2 = 1^2$

$x^2 + b^2 = 1$

$b^2 = 1 - x^2$

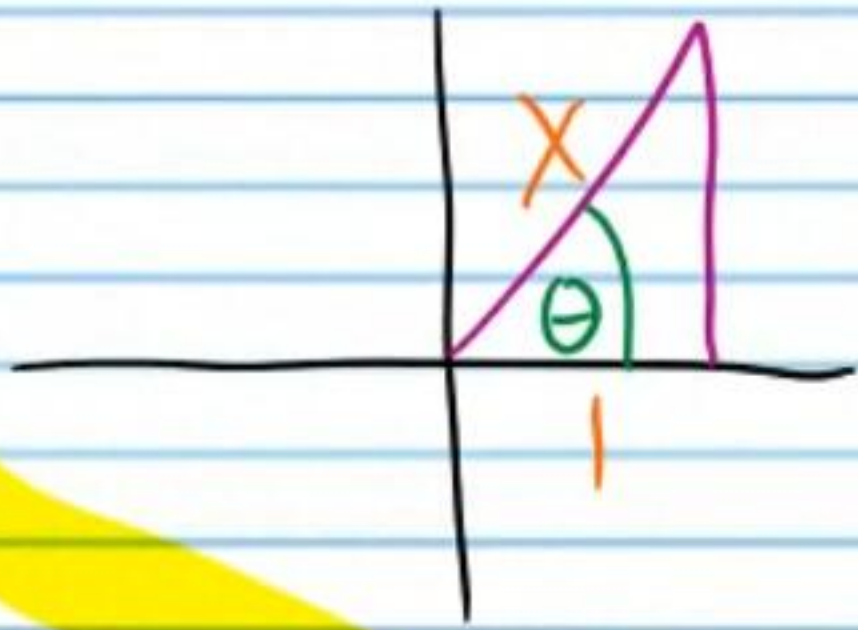
$b = \pm \sqrt{1-x^2}$

$b = \sqrt{1-x^2}$

②⑦ $\cos(\sec^{-1} x)$

I) $\theta = \sec^{-1} x$

$\sec \theta = x = \frac{x}{1}$



II) $\cos(\sec^{-1} x)$

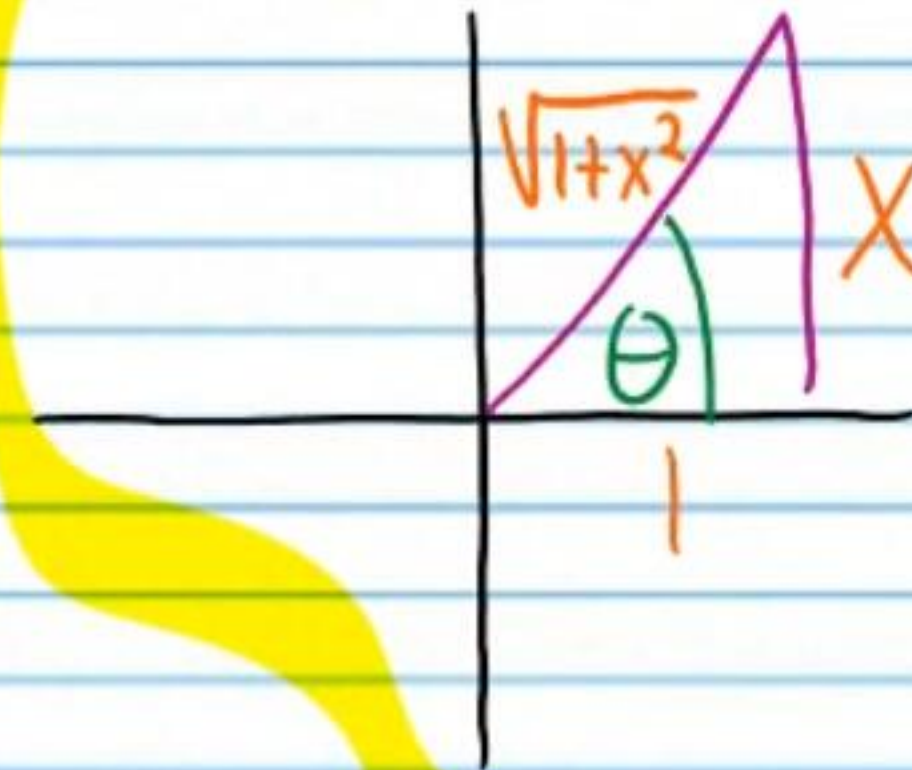
$\cos \theta$

$\boxed{\frac{1}{x}}$

②⑧ $\sec(\cot^{-1} \frac{1}{x})$

I) $\theta = \cot^{-1} \frac{1}{x}$

$\cot \theta = \frac{1}{x}$



II) $\sec(\cot^{-1} \frac{1}{x})$

$\sec \theta$

$\frac{\sqrt{1+x^2}}{1}$

$\boxed{\sqrt{1+x^2}}$

$a^2 + b^2 = c^2$

$1^2 + x^2 = c^2$

$1 + x^2 = c^2$

$\pm \sqrt{1+x^2} = c$

$c = \sqrt{1+x^2}$